

LAB PROGRAMS

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Subject: Data Handling and Visualization

Code: DSA0604

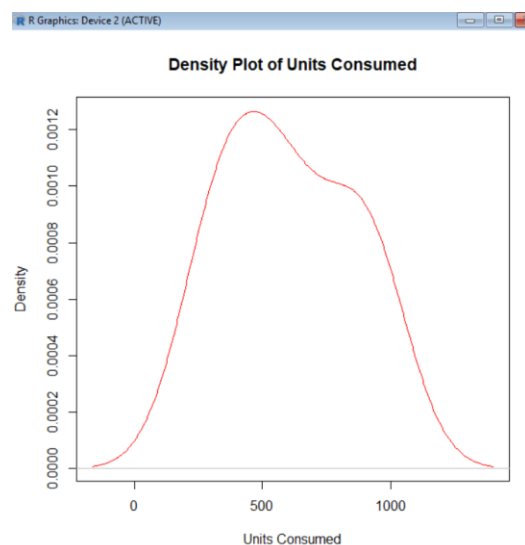
SET 21 – Energy Consumption

Q1. Histogram and Density Plot

CODE:

```
data <- data.frame(  
Units=c(320,540,880,350,610,920)  
)  
  
hist(data$Units,  
      main="Histogram of Units Consumed",  
      xlab="Units Consumed",  
      col="skyblue")  
  
plot(density(data$Units),  
      main="Density Plot of Units Consumed",  
      xlab="Units Consumed",  
      col="red")
```

OUTPUT:

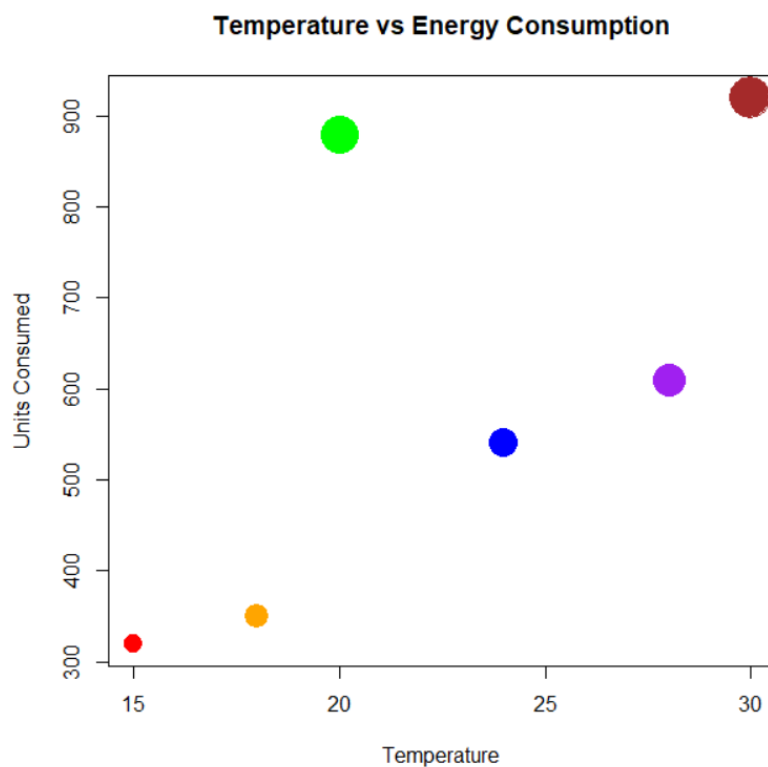


Q2. Temperature vs Units Consumed – Bubble Plot

CODE:

```
temperature <- c(15,24,20,18,28,30)
units <- c(320,540,880,350,610,920)
peak <- c(4,6,8,5,7,9)
colours <- c("red","blue","green","orange","purple","brown")
plot(temperature, units,
     cex=peak/2,
     pch=19,
     col=colours,
     xlab="Temperature",
     ylab="Units Consumed",
     main="Temperature vs Energy Consumption")
```

OUTPUT:



Q3. Average Renewable Usage by Sector

CODE:

```
sector <- c("Residential","Commercial","Industrial",
```

```

"Residential","Commercial","Industrial")

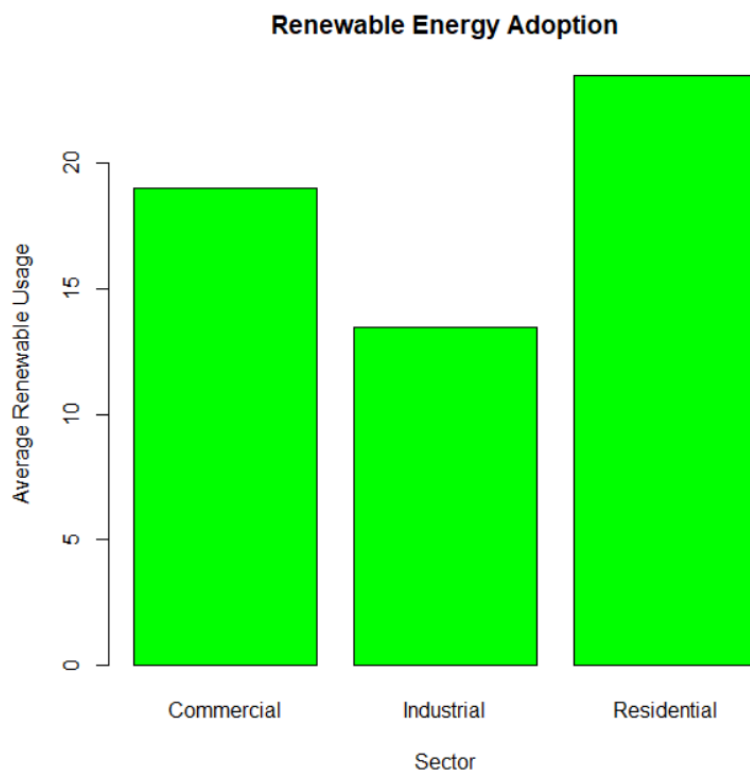
renewable <- c(22,18,12,25,20,15)

avg <- aggregate(renewable,
                 by=list(sector),
                 FUN=mean)

barplot(avg$x,
        names.arg=avg$Group.1,
        col="green",
        xlab="Sector",
        ylab="Average Renewable Usage",
        main="Renewable Energy Adoption")

```

OUTPUT:



Q4. Monthly Energy Usage + Heatmap

CODE:

```

sector <- c("Residential","Commercial","Industrial",
            "Residential","Commercial","Industrial")

region <- c("North","South","West","East","North","South")

```

```
month <- c("Jan","Jan","Feb","Feb","Mar","Mar")
```

```
units <- c(320,540,880,350,610,920)
```

```
monthly <- aggregate(units,
```

```
  by=list(month),
```

```
  FUN=sum)
```

```
plot(monthly$x,
```

```
  type="o",
```

```
  pch=19,
```

```
  xaxt="n",
```

```
  xlab="Month",
```

```
  ylab="Units Consumed",
```

```
  main="Monthly Energy Usage")
```

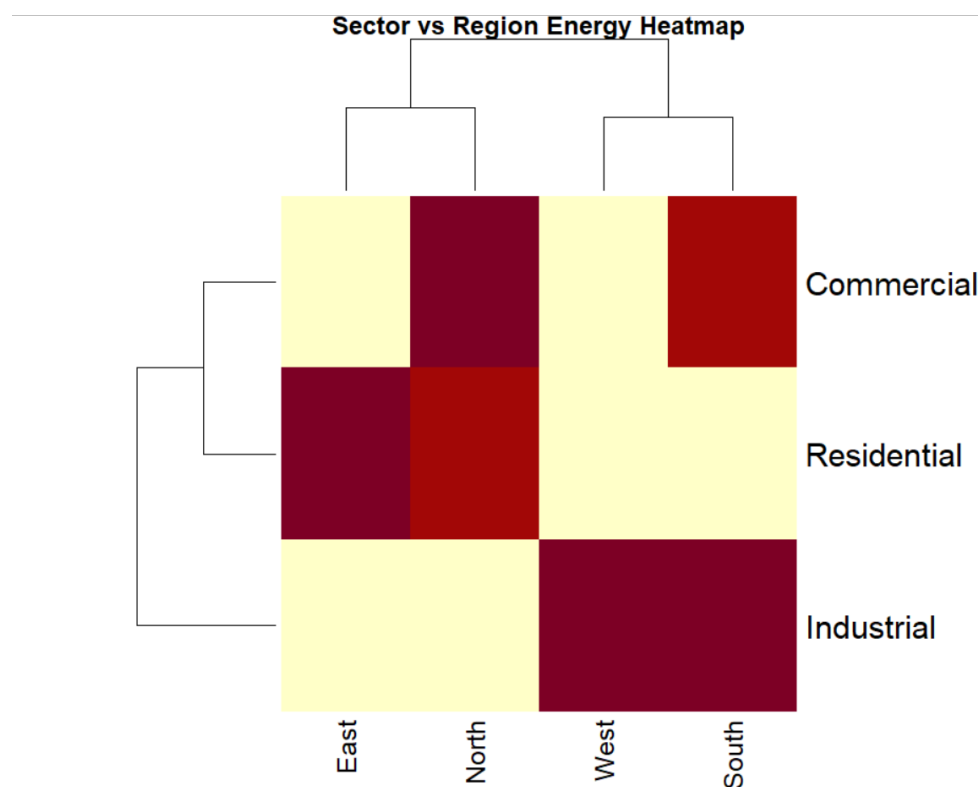
```
axis(1,1:length(monthly$x),monthly$Group.1)
```

```
heat <- xtabs(units ~ sector + region)
```

```
heatmap(heat,
```

```
  main="Sector vs Region Energy Heatmap")
```

OUTPUT:



SET 22 – Time Series Analysis

Q1. Time Series Line Chart

CODE:

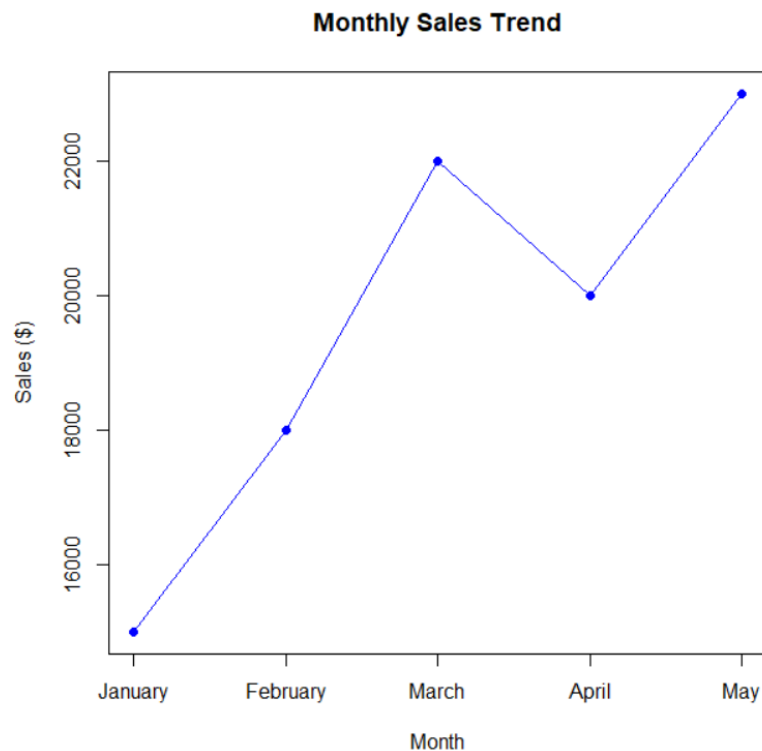
```
month <- c("January","February","March","April","May")
```

```
sales <- c(15000,18000,22000,20000,23000)
```

```
plot(sales,  
     type="o",  
     pch=19,  
     col="blue",  
     xaxt="n",  
     xlab="Month",  
     ylab="Sales ($)",  
     main="Monthly Sales Trend")
```

```
axis(1, 1:5, month)
```

OUTPUT:



Q2. Scatter Plot – Advertising Budget vs Sales

CODE:

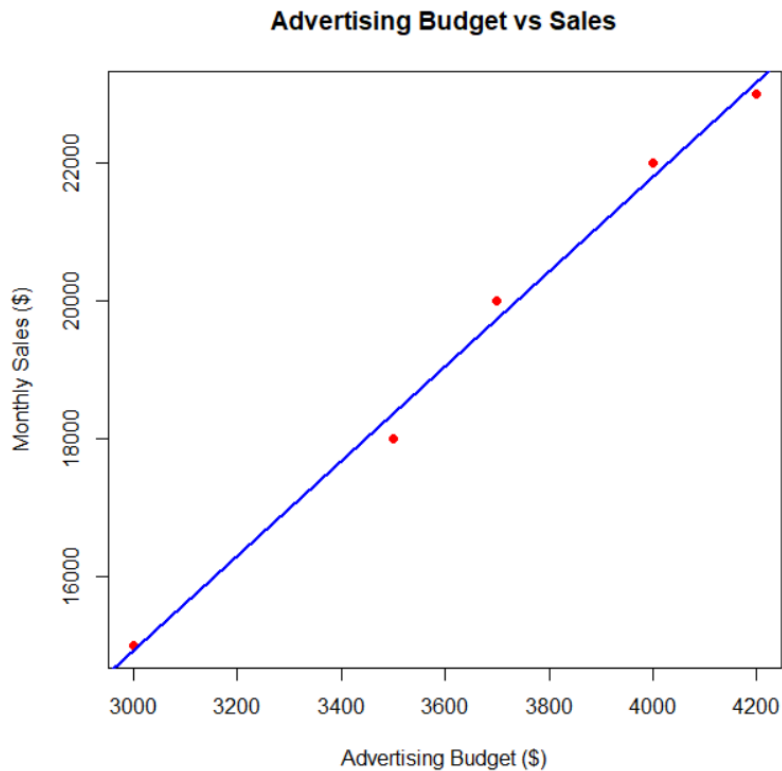
```
advertising <- c(3000,3500,4000,3700,4200)
```

```
sales <- c(15000,18000,22000,20000,23000)
```

```
plot(advertising,  
     sales,  
     pch=19,  
     col="red",  
     xlab="Advertising Budget ($)",  
     ylab="Monthly Sales ($)",  
     main="Advertising Budget vs Sales")
```

```
abline(lm(sales ~ advertising),  
       col="blue",  
       lwd=2)
```

OUTPUT:



Q3. Line Chart + Pie Chart in R

CODE:

```
month <- c("January","February","March","April","May")
```

```
sales <- c(15000,18000,22000,20000,23000)
```

```
par(mfrow=c(1,2))
```

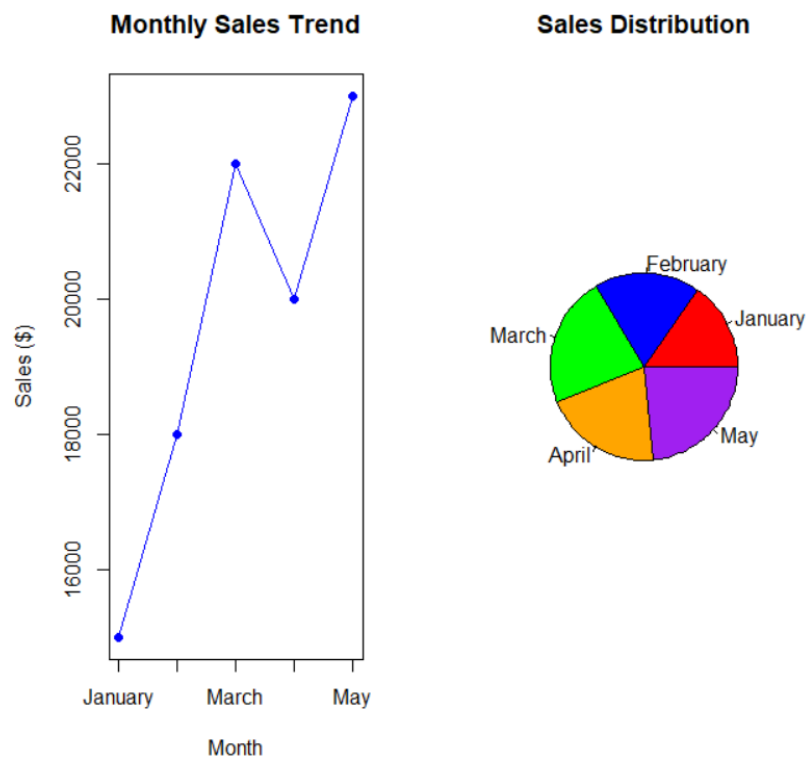
```
plot(sales,  
     type="o",  
     pch=19,  
     col="blue",  
     xaxt="n",  
     xlab="Month",  
     ylab="Sales ($)",  
     main="Monthly Sales Trend")
```

```
axis(1,1:5,month)
```

```
pie(sales,  
    labels=month,  
    main="Sales Distribution",  
    col=c("red","blue","green","orange","purple"))
```

```
par(mfrow=c(1,1))
```

OUTPUT:

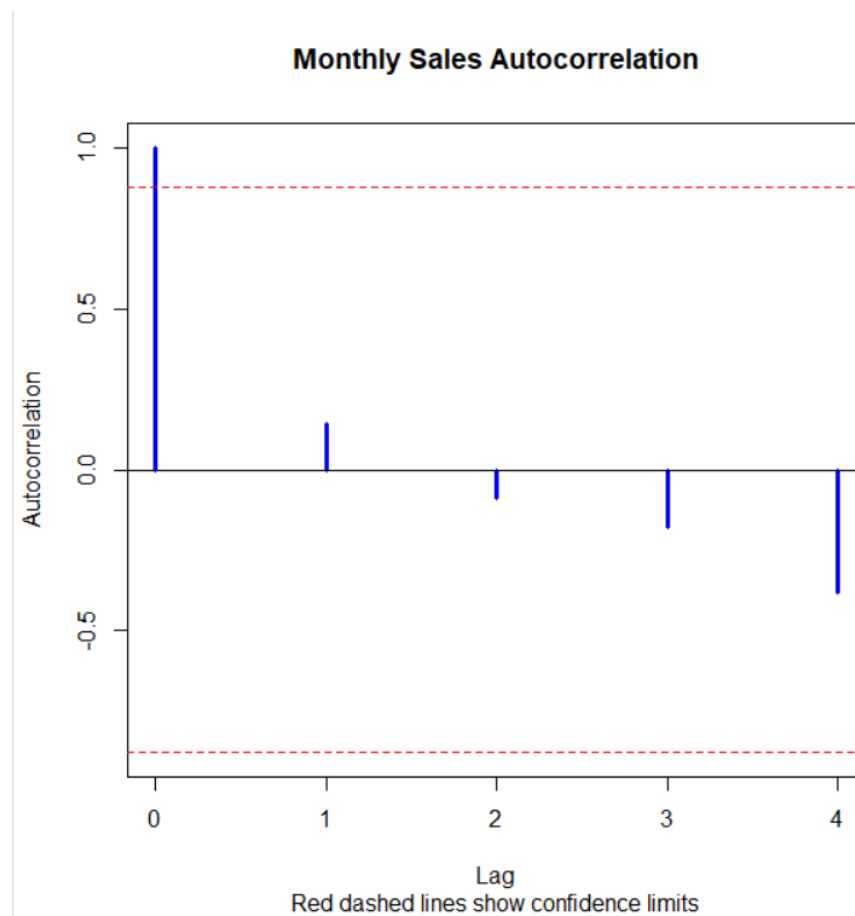


Q4. Autocorrelation Plot

CODE:

```
sales <- c(15000, 18000, 22000, 20000, 23000)
sales_ts <- ts(sales)
acf(sales_ts,
    lag.max=4,
    main="Monthly Sales Autocorrelation",
    xlab="Lag",
    ylab="Autocorrelation",
    col="blue",
    lwd=3,
    ci.col="red",
    ci.lwd=2)
title(sub="Red dashed lines show confidence limits")
```

OUTPUT:



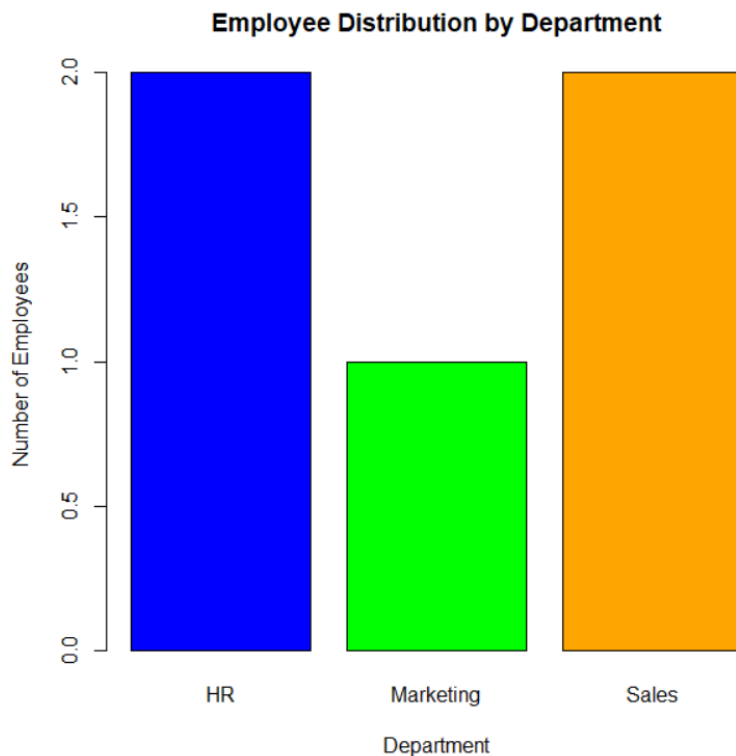
SET 23 – Employee Performance Analysis

Q1. Bar Chart – Department Distribution

CODE:

```
department <- c("Sales","HR","Marketing","Sales","HR")
count <- table(department)
barplot(count,
        col=c("blue","green","orange"),
        main="Employee Distribution by Department",
        xlab="Department",
        ylab="Number of Employees")
```

OUTPUT:



Q2. Line Chart – Employee Performance Trend

CODE:

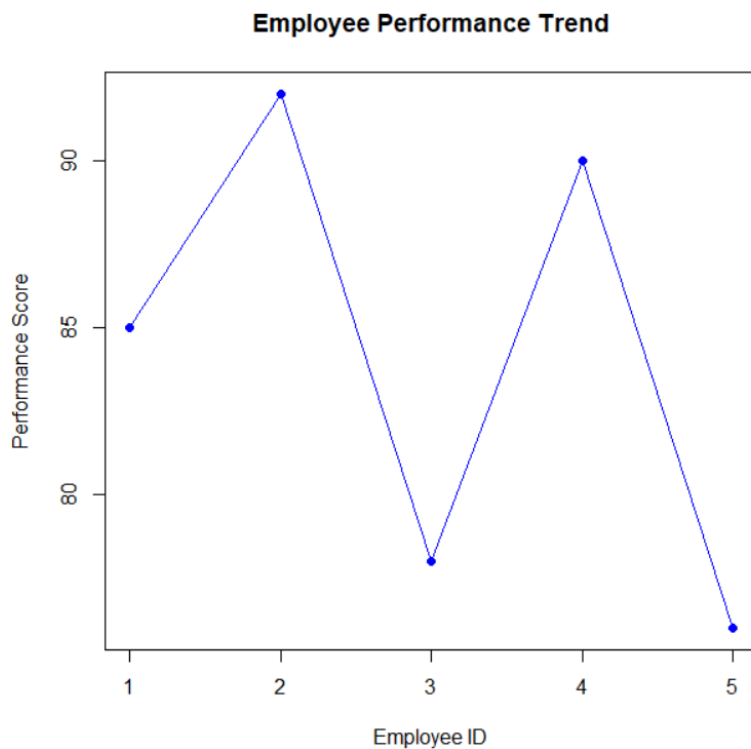
```
employee_id <- c(1,2,3,4,5)
performance <- c(85,92,78,90,76)
plot(employee_id,
      performance,
```

```

type="o",
pch=19,
col="blue",
xlab="Employee ID",
ylab="Performance Score",
main="Employee Performance Trend")

```

OUTPUT:



Q3. Dashboard in R – Bar Chart + Scatter Plot

CODE:

```

department <- c("Sales","HR","Marketing","Sales","HR")
years <- c(5,3,7,4,2)
performance <- c(85,92,78,90,76)
count <- table(department)
par(mfrow=c(1,2))
barplot(count,
        col=c("blue","green","orange"),
        main="Department Distribution",
        xlab="Department",

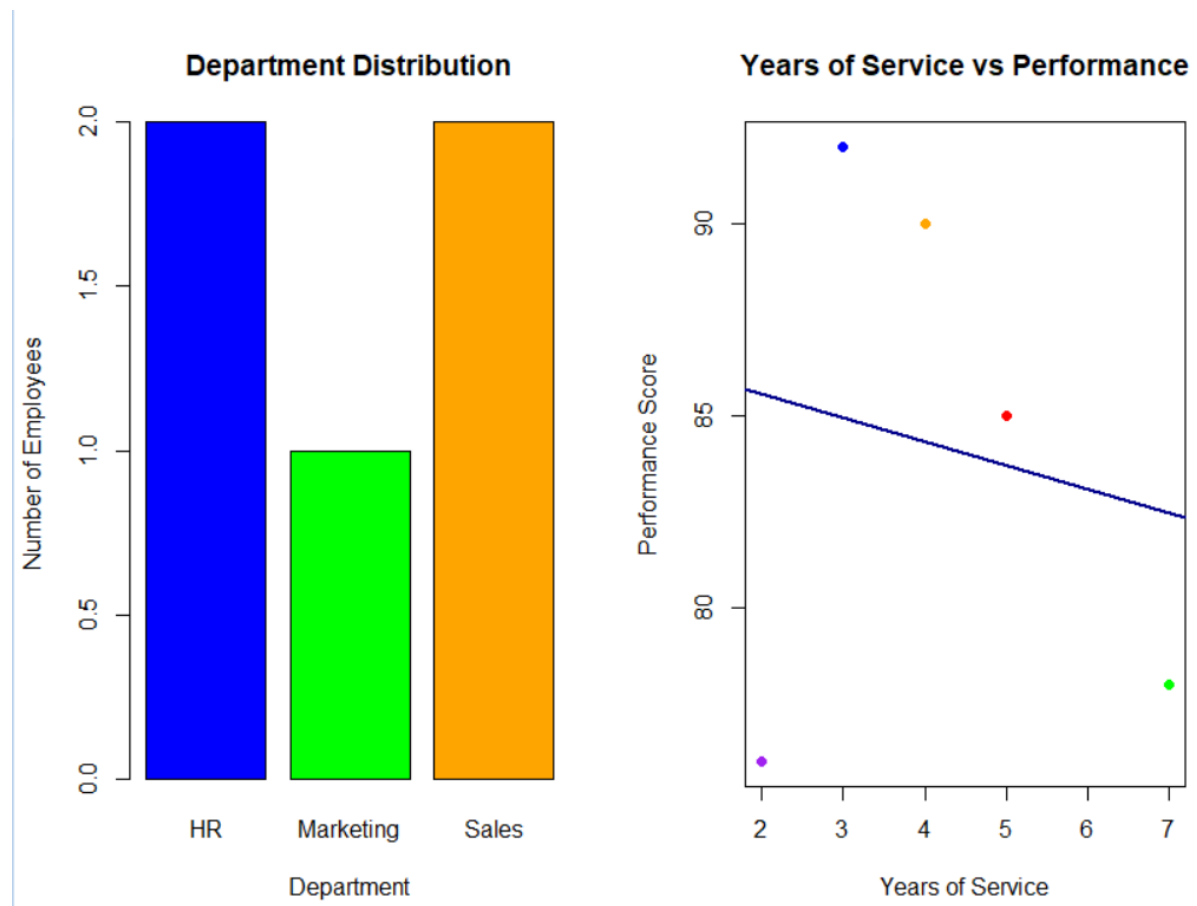
```

```

ylab="Number of Employees")
plot(years,
      performance,
      pch=19,
      col=c("red","blue","green","orange","purple"),
      xlab="Years of Service",
      ylab="Performance Score",
      main="Years of Service vs Performance")
abline(lm(performance ~ years),
      col="darkblue",
      lwd=2)
par(mfrow=c(1,1))

```

OUTPUT:



Q4. Performance Data Table – Colored

CODE:

```

employee <- data.frame(
  Employee_ID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  Years_of_Service = c(5,3,7,4,2),
  Performance_Score = c(85,92,78,90,76)
)

library(gridExtra)
library(grid)

grid.table(employee,
  rows = NULL,
  theme = ttheme_default(
    core = list(
      bg_params = list(
        fill = c("lightblue",
                  "lightgreen",
                  "lightyellow",
                  "lightpink",
                  "lightgray")
      ),
      fg_params = list(
        col = "black",
        fontsize = 12
      )
    ),
    colhead = list(
      bg_params = list(fill = "steelblue"),
      fg_params = list(
        col = "white",
        fontsize = 13
      )
    )
  )
)

```

))

OUTPUT:

Employee_ID	Department	Years_of_Service	Performance_Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

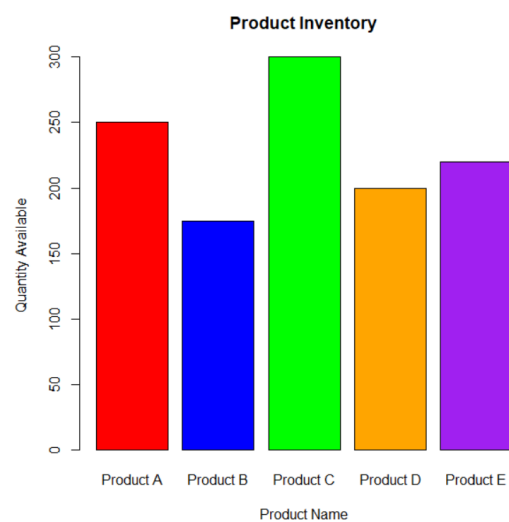
SET 24 – Product Inventory Management

Q1. Bar Chart – Quantity of Each Product

CODE:

```
product <- c("Product A","Product B","Product C",  
            "Product D","Product E")  
quantity <- c(250,175,300,200,220)  
barplot(quantity,  
        names.arg=product,  
        col=c("red","blue","green","orange","purple"),  
        main="Product Inventory",  
        xlab="Product Name",  
        ylab="Quantity Available")
```

OUTPUT:



Q2. Stacked Bar Chart – Product Categories

CODE:

```
product <- c("A","B","C","D","E")
```

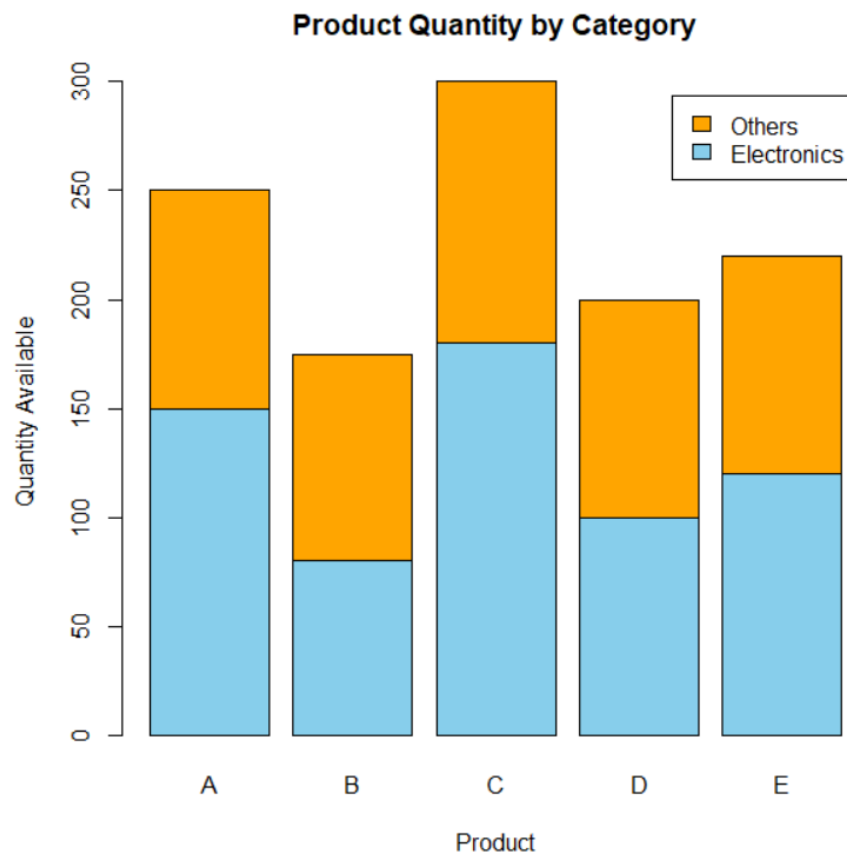
```
Electronics <- c(150,80,180,100,120)
```

```
Others <- c(100,95,120,100,100)
```

```
data <- rbind(Electronics, Others)
```

```
barplot(data,  
  names.arg=product,  
  col=c("skyblue","orange"),  
  main="Product Quantity by Category",  
  xlab="Product",  
  ylab="Quantity Available",  
  legend=rownames(data))
```

OUTPUT:



Q3. Dashboard in R – Bar Chart + Stacked Bar Chart

CODE:

```
product <- c("A","B","C","D","E")
quantity <- c(250,175,300,200,220)
Electronics <- c(150,80,180,100,120)
Others <- c(100,95,120,100,100)
data <- rbind(Electronics, Others)
par(mfrow=c(2,2),
    mar=c(4,4,3,2))
total <- sum(quantity)
plot.new()
text(0.5,0.65,
     "TOTAL INVENTORY",
     cex=1.5,
     font=2)
text(0.5,0.40,
     total,
     cex=3,
     col="blue",
     font=2)
barplot(quantity,
        names.arg=product,
        col=c("red","blue","green","orange","purple"),
        main="Quantity by Product",
        xlab="Product",
        ylab="Quantity")
barplot(data,
        names.arg=product,
        col=c("skyblue","orange"),
        main="Inventory by Category",
        xlab="Product",
```

```

        ylab="Quantity",
        legend=rownames(data))
plot.new()
text(0.5,0.85,
     "INVENTORY SUMMARY",
     cex=1.4,
     font=2)
text(0.5,0.65,
     paste("Highest: Product", product[which.max(quantity)]),
     cex=1.2)
text(0.5,0.50,
     paste("Quantity:", max(quantity)),
     cex=1.2)
text(0.5,0.35,
     paste("Lowest: Product", product[which.min(quantity)]),
     cex=1.2)
text(0.5,0.20,
     paste("Quantity:", min(quantity)),
     cex=1.2)

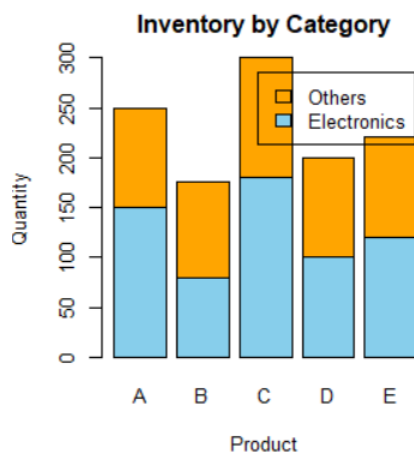
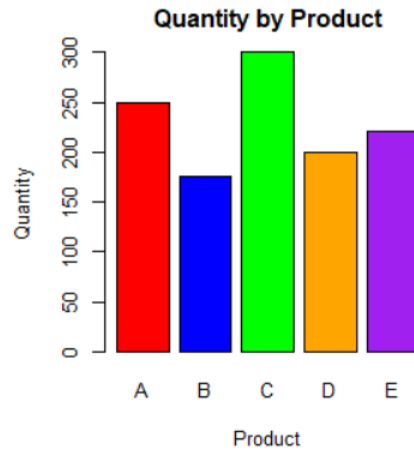
mtext("PRODUCT INVENTORY MANAGEMENT DASHBOARD",
      outer=TRUE,
      cex=1.5,
      font=2)
par(mfrow=c(1,1))

```

OUTPUT:

TOTAL INVENTORY

1145



INVENTORY SUMMARY

Highest: Product C

Quantity: 300

Lowest: Product B

Quantity: 175

Q4. Scatter Plot – Price vs Quantity

CODE:

```
product <- c("Product A","Product B","Product C",
```

```
            "Product D","Product E")
```

```
price <- c(100,80,150,120,90)
```

```
quantity <- c(250,175,300,200,220)
```

```
plot(price,
```

```
     quantity,
```

```
     pch=19,
```

```
     cex=2,
```

```
     col=c("red","blue","green","orange","purple"),
```

```
     xlab="Product Price",
```

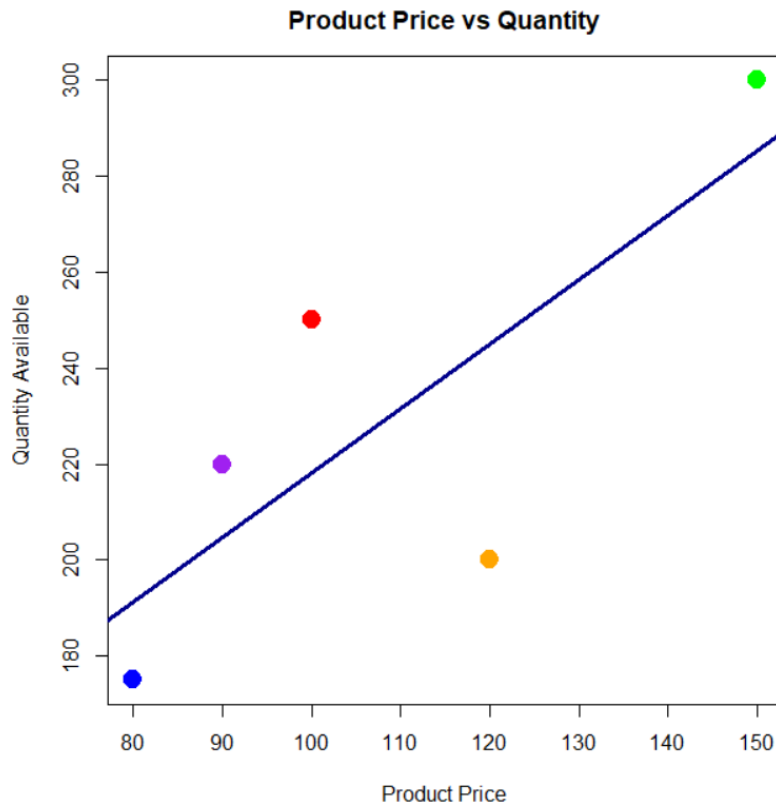
```
     ylab="Quantity Available",
```

```
     main="Product Price vs Quantity")
```

```
abline(lm(quantity ~ price),
```

```
col="darkblue",  
lwd=3)
```

OUTPUT:



SET 25 – Website Traffic Analysis

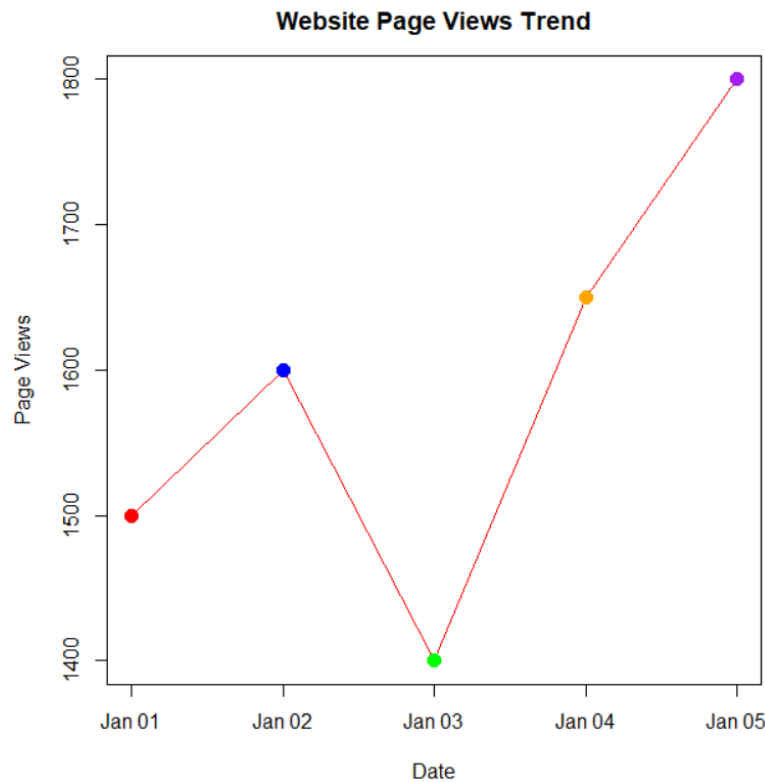
Q1. Line Chart – Page Views Over Time

CODE:

```
date <- as.Date(c("2023-01-01",  
                  "2023-01-02",  
                  "2023-01-03",  
                  "2023-01-04",  
                  "2023-01-05"))  
  
page_views <- c(1500,1600,1400,1650,1800)  
  
plot(date,  
      page_views,  
      type="o",  
      pch=19,  
      cex=1.5,
```

```
col=c("red","blue","green","orange","purple"),
xlab="Date",
ylab="Page Views",
main="Website Page Views Trend")
```

OUTPUT:

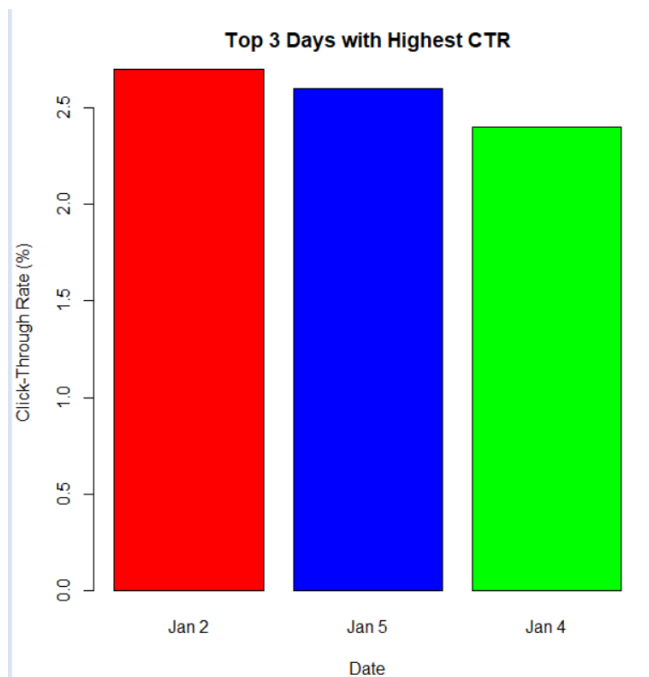


Q2. Top N Days with Highest Click-Through Rate

CODE:

```
date <- c("Jan 1","Jan 2","Jan 3","Jan 4","Jan 5")
ctr <- c(2.3,2.7,2.0,2.4,2.6)
top <- order(ctr, decreasing=TRUE)[1:3]
barplot(ctr[top],
        names.arg=date[top],
        col=c("red","blue","green"),
        main="Top 3 Days with Highest CTR",
        xlab="Date",
        ylab="Click-Through Rate (%)")
```

OUTPUT:



Q3. Stacked Area Chart – User Interactions

CODE:

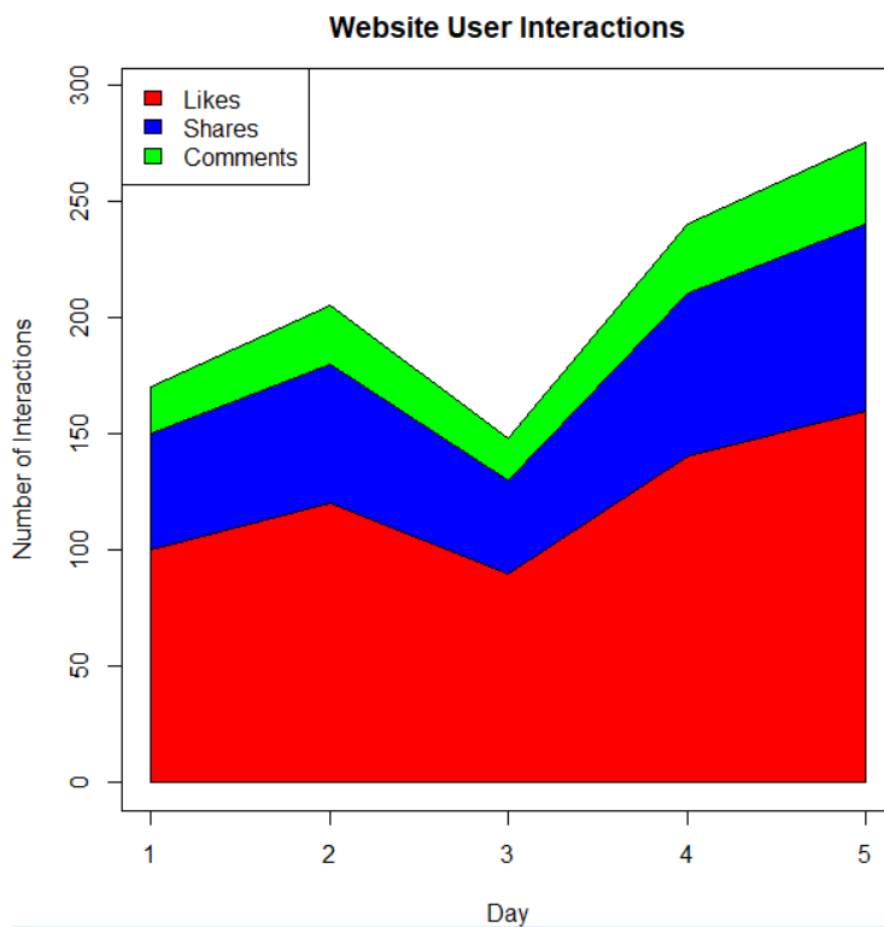
```
days <- 1:5
likes <- c(100,120,90,140,160)
shares <- c(50,60,40,70,80)
comments <- c(20,25,18,30,35)
total <- likes + shares + comments
plot(days,
      total,
      type="n",
      ylim=c(0,max(total)+20),
      xlab="Day",
      ylab="Number of Interactions",
      main="Website User Interactions")
polygon(c(days,rev(days)),
        c(rep(0,5),rev(likes)),
        col="red")
polygon(c(days,rev(days)),
        c(likes,rev(likes+shares)),
```

```

col="blue")
polygon(c(days,rev(days)),
       c(likes+shares,
         rev(likes+shares+comments))),
col="green")
legend("topleft",
      legend=c("Likes","Shares","Comments"),
      fill=c("red","blue","green"))

```

OUTPUT:



Q4. Website Traffic Dashboard in R

CODE:

```

source <- c("Google", "Facebook", "YouTube", "Direct", "Instagram")
views <- c(2500, 1800, 1500, 2200, 1300)
par(mfrow=c(2,2))
pie(views,

```

```

labels=source,
main="Traffic Sources",
col=c("red","blue","green","orange","purple"))
barplot(views,
        names.arg=source,
        col=c("red","blue","green","orange","purple"),
        main="Page Views by Source",
        xlab="Source",
        ylab="Page Views",
        las=2)
plot(views,
     type="o",
     pch=19,
     lwd=3,
     col="blue",
     xaxt="n",
     xlab="Traffic Source",
     ylab="Page Views",
     main="Traffic Trend")
axis(1, 1:5, source)
plot.new()
text(0.5,0.85,
     "TRAFFIC SUMMARY",
     cex=1.5,
     font=2)
text(0.5,0.65,
     paste("Total Views =", sum(views)),
     cex=1.2)
text(0.5,0.50,
     paste("Highest =", source[which.max(views)]),
     cex=1.2)

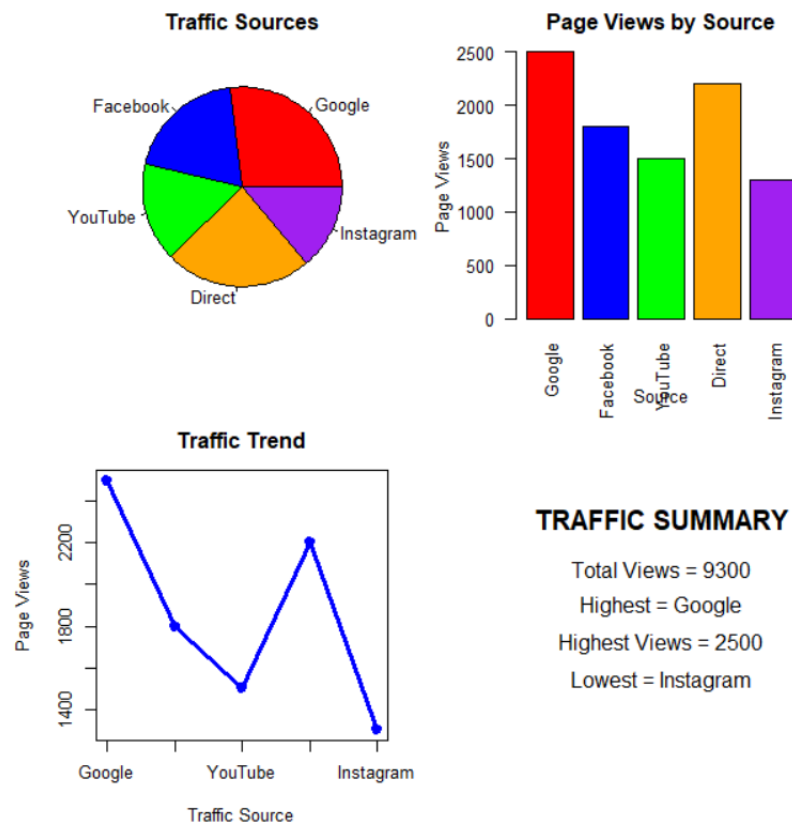
```

```

text(0.5,0.35,
     paste("Highest Views =", max(views)),
     cex=1.2)
text(0.5,0.20,
     paste("Lowest =", source[which.min(views)]),
     cex=1.2)
par(mfrow=c(1,1))

```

OUTPUT:



SET 26 – Student Mini Data

Q1. Histogram + Boxplot by Gender

CODE:

```

student <- data.frame(
  Student_ID = c("S01", "S02", "S03", "S04", "S05", "S06"),
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
  Study_Hours = c(2.0, 3.5, 1.5, 4.0, 2.8, 3.0),
  Attendance = c(78, 90, 70, 95, 85, 92),
  Math_Score = c(62, 80, 55, 90, 72, 82),

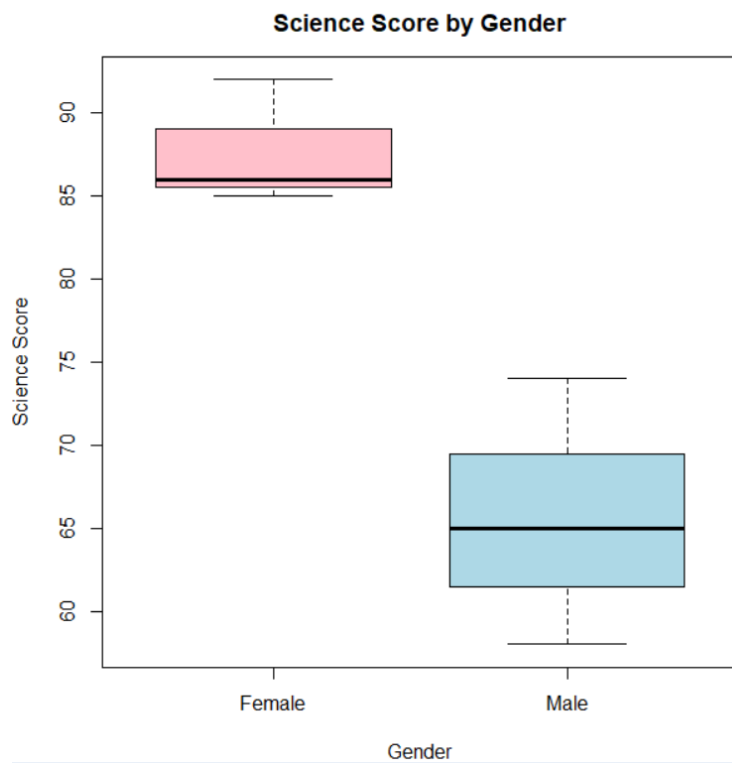
```

```

Science_Score = c(65,85,58,92,74,86),
Exam_Date = as.Date(c("2025-01-10","2025-01-10",
                      "2025-02-12","2025-02-12",
                      "2025-03-15","2025-03-15"))
)
hist(student$Math_Score,
      col="skyblue",
      main="Distribution of Math Scores",
      xlab="Math Score",
      ylab="Number of Students")
boxplot(Science_Score ~ Gender,
        data=student,
        col=c("pink","lightblue"),
        main="Science Score by Gender",
        xlab="Gender",
        ylab="Science Score")

```

OUTPUT:

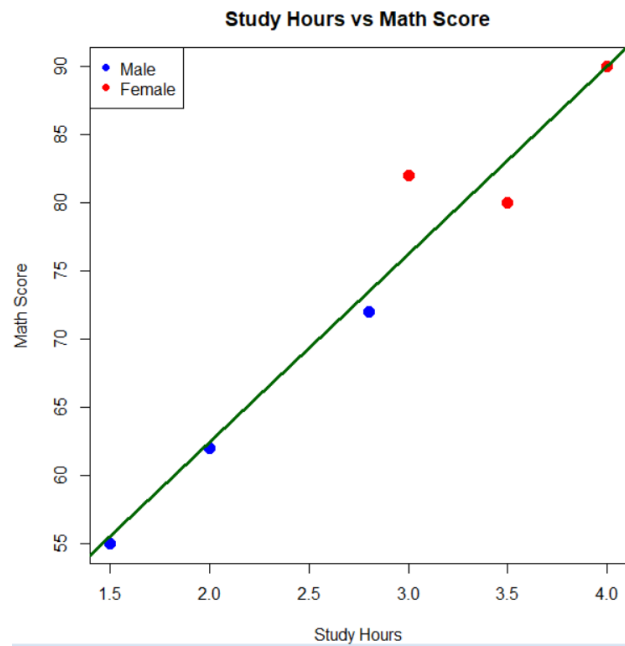


Q2. Study Hours vs Math Score

CODE:

```
student <- data.frame(  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Study_Hours = c(2.0,3.5,1.5,4.0,2.8,3.0),  
  Math_Score = c(62,80,55,90,72,82)  
)  
student$color <- ifelse(student$Gender=="Male",  
  "blue","red")  
plot(student$Study_Hours,  
  student$Math_Score,  
  pch=19,  
  cex=1.5,  
  col=student$color,  
  xlab="Study Hours",  
  ylab="Math Score",  
  main="Study Hours vs Math Score")  
abline(lm(Math_Score ~ Study_Hours,  
  data=student),  
  col="darkgreen",  
  lwd=3)  
legend("topleft",  
  legend=c("Male","Female"),  
  col=c("blue","red"),  
  pch=19)
```

OUTPUT:



Q3. Monthly Average Math Score + Moving Average

CODE:

```
student <- data.frame(
  Math_Score = c(62,80,55,90,72,82),
  Exam_Date = as.Date(c("2025-01-10","2025-01-10",
    "2025-02-12","2025-02-12",
    "2025-03-15","2025-03-15"))
)
student$Month <- format(student$Exam_Date, "%Y-%m")
monthly <- aggregate(Math_Score ~ Month,
  data=student,
  mean)
print(monthly)
moving_avg <- c(
  NA,
  mean(monthly$Math_Score[1:2]),
  mean(monthly$Math_Score[2:3])
)
plot(monthly$Math_Score,
  type="o",
```

```

pch=19,

col="blue",

xaxt="n",

xlab="Month",

ylab="Average Math Score",

main="Monthly Average Math Score")

axis(1,

1:nrow(monthly),

monthly$Month)

lines(moving_avg,

type="o",

pch=17,

col="red",

lwd=3)

legend("topleft",

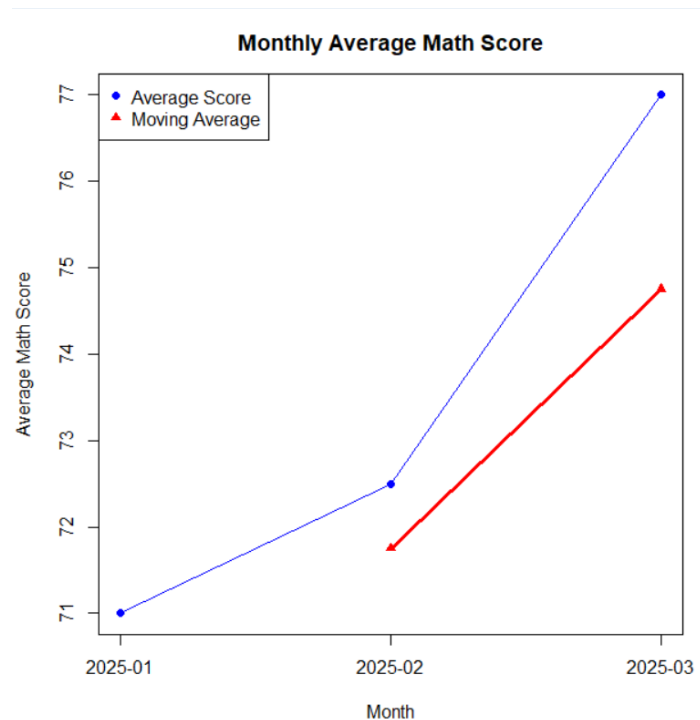
legend=c("Average Score","Moving Average"),

col=c("blue","red"),

pch=c(19,17))

```

OUTPUT:



Q4. R Dashboard – Gender Average + Score Trend

CODE:

```
student <- data.frame(
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Math_Score = c(62,80,55,90,72,82),
  Exam_Date = as.Date(c("2025-01-10","2025-01-10",
    "2025-02-12","2025-02-12",
    "2025-03-15","2025-03-15"))
)
male_avg <- mean(student$Math_Score[student$Gender=="Male"])
female_avg <- mean(student$Math_Score[student$Gender=="Female"])
student$Month <- format(student$Exam_Date,"%Y-%m")
monthly <- aggregate(Math_Score ~ Month,
  data=student,
  mean)
par(mfrow=c(2,2),
  mar=c(5,4,4,2))
barplot(c(male_avg,female_avg),
  names.arg=c("Male","Female"),
  col=c("blue","pink"),
  main="Math Score by Gender",
  xlab="Gender",
  ylab="Average Score")
plot(monthly$Math_Score,
  type="o",
  pch=19,
  col="darkgreen",
  xaxt="n",
  main="Monthly Math Trend",
  xlab="Month",
  ylab="Average Score")
```

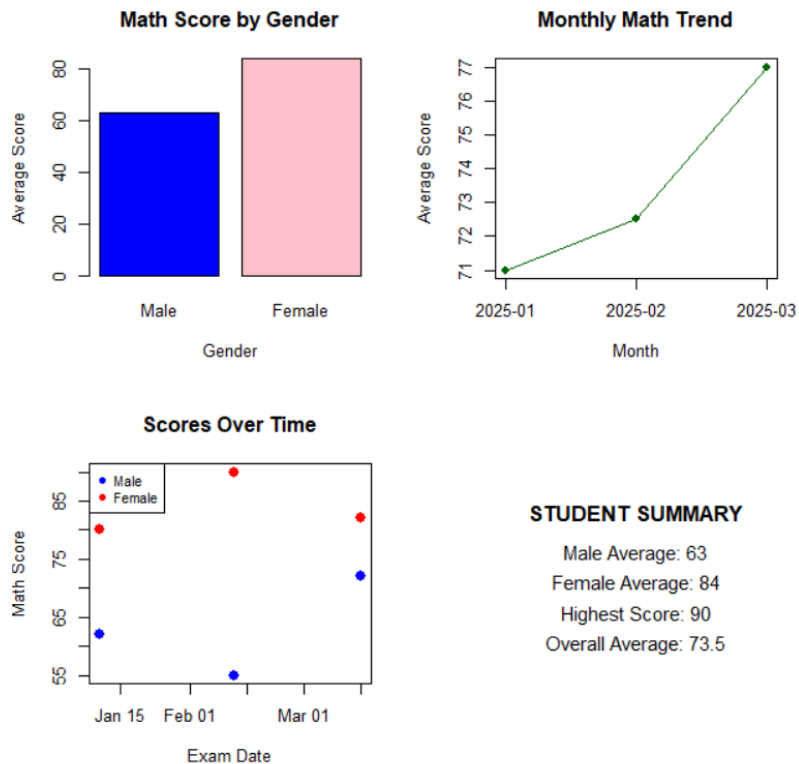
```

axis(1,
      at=1:length(monthly$Month),
      labels=monthly$Month)
plot(student$Exam_Date,
      student$Math_Score,
      pch=19,
      cex=1.3,
      col=ifelse(student$Gender=="Male",
                  "blue","red"),
      main="Scores Over Time",
      xlab="Exam Date",
      ylab="Math Score")
legend("topleft",
      legend=c("Male","Female"),
      col=c("blue","red"),
      pch=19,
      cex=0.8)
plot.new()
text(0.5,0.80,
      "STUDENT SUMMARY",
      cex=1.3,
      font=2)
text(0.5,0.60,
      paste("Male Average:",round(male_avg,1)),
      cex=1.1)
text(0.5,0.45,
      paste("Female Average:",round(female_avg,1)),
      cex=1.1)
text(0.5,0.30,
      paste("Highest Score:",max(student$Math_Score)),
      cex=1.1)

```

```
text(0.5,0.15,
     paste("Overall Average:",
           round(mean(student$Math_Score),1)),
     cex=1.1)
par(mfrow=c(1,1))
```

OUTPUT:



SET 27 – Patient Health Risk Analysis

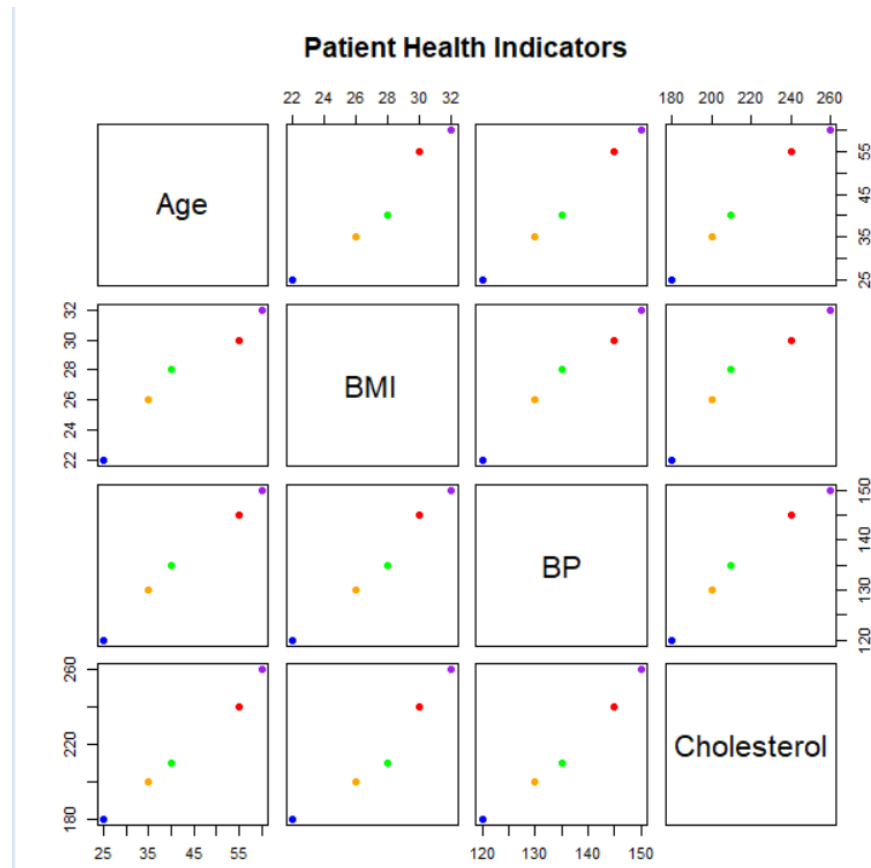
Q1. Scatterplot Matrix

CODE:

```
patient <- data.frame(
  Age = c(25,40,55,35,60),
  BMI = c(22,28,30,26,32),
  BP = c(120,135,145,130,150),
  Cholesterol = c(180,210,240,200,260)
)
pairs(patient,
      main="Patient Health Indicators",
```

```
pch=19,
col=c("blue","green","red","orange","purple"))
```

OUTPUT:



Q2. Q-Q Plot and ECDF for Cholesterol

CODE:

```
cholesterol <- c(180,210,240,200,260)
```

```
par(mfrow=c(1,2))
```

```
qqnorm(cholesterol,
```

```
  pch=19,
```

```
  col="blue",
```

```
  main="Q-Q Plot of Cholesterol")
```

```
qqline(cholesterol,
```

```
  col="red",
```

```
  lwd=2)
```

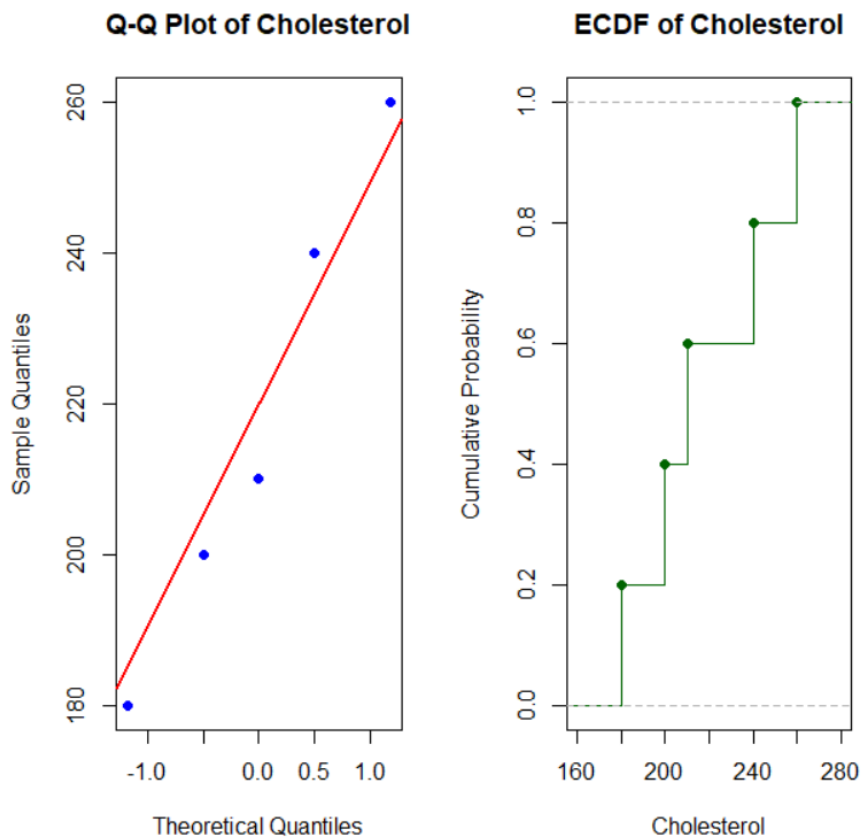
```
plot(ecdf(cholesterol),
```

```

verticals=TRUE,
do.points=TRUE,
pch=19,
col="darkgreen",
main="ECDF of Cholesterol",
xlab="Cholesterol",
ylab="Cumulative Probability")
par(mfrow=c(1,1))

```

OUTPUT:



Q3. Average Health Indicators – Bar Chart

CODE;

```

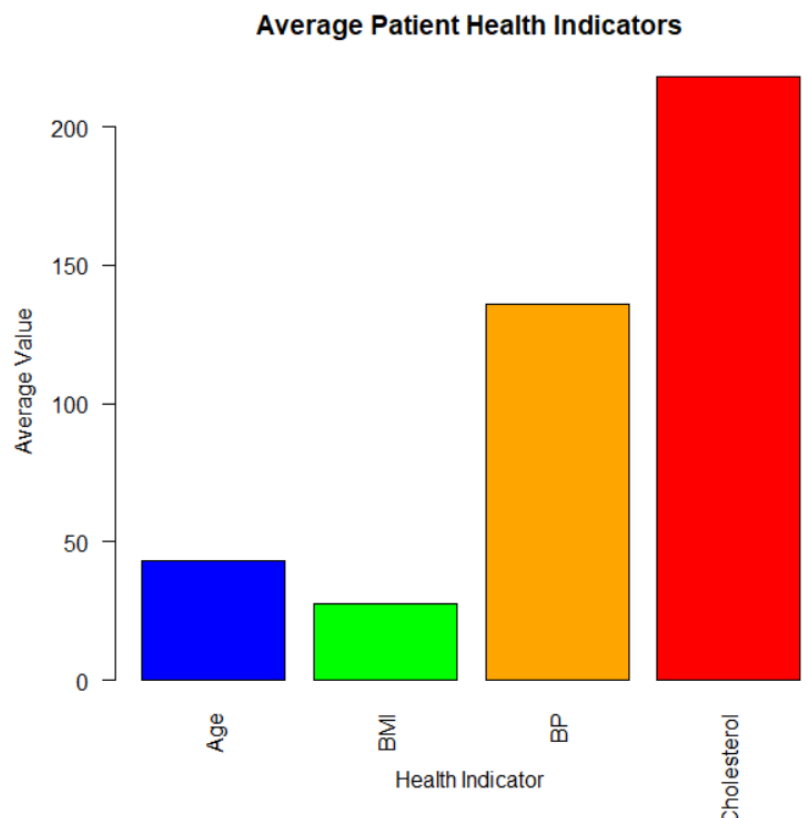
patient <- data.frame(
  Age = c(25,40,55,35,60),
  BMI = c(22,28,30,26,32),
  BP = c(120,135,145,130,150),
  Cholesterol = c(180,210,240,200,260)
)

```



```
)  
average <- c(  
  mean(patient$Age),  
  mean(patient$BMI),  
  mean(patient$BP),  
  mean(patient$Cholesterol)  
)  
barplot(average,  
  names.arg=c("Age","BMI","BP","Cholesterol"),  
  col=c("blue","green","orange","red"),  
  main="Average Patient Health Indicators",  
  xlab="Health Indicator",  
  ylab="Average Value",  
  las=2)
```

OUTPUT:



Q4. R Dashboard – Heatmap + Age Risk Analysis

CODE:

```
patient <- data.frame(
  Patient_ID=c("P1","P2","P3","P4","P5"),
  Age=c(25,40,55,35,60),
  BMI=c(22,28,30,26,32),
  BP=c(120,135,145,130,150),
  Cholesterol=c(180,210,240,200,260)
)

patient$Risk_Score <-
  scale(patient$BMI)[,1] +
  scale(patient$BP)[,1] +
  scale(patient$Cholesterol)[,1]

par(mfrow=c(2,2),
    mar=c(5,4,4,2))

health <- patient[,c("Age","BMI","BP","Cholesterol")]

heatmap(as.matrix(health),
  Rowv=NA,
  Colv=NA,
  scale="column",
  col=heat.colors(20),
  main="Patient Health Heatmap",
  xlab="Health Indicators",
  ylab="Patients")

plot(patient$Age,
  patient$Risk_Score,
  pch=19,
  cex=1.5,
  col=c("green","blue","orange","red","darkred"),
  xlab="Age",
  ylab="Risk Score",
```

```

    main="Age vs Health Risk")
text(patient$Age,
      patient$Risk_Score,
      labels=patient$Patient_ID,
      pos=3,
      cex=0.8)
barplot(patient$Risk_Score,
        names.arg=patient$Patient_ID,
        col=c("green","blue","orange","red","darkred"),
        main="Patient Risk Score",
        xlab="Patient",
        ylab="Risk Score")
plot.new()
text(0.5,0.85,
     "PATIENT RISK SUMMARY",
     cex=1.3,
     font=2)
highest <- patient$Patient_ID[
  which.max(patient$Risk_Score)
]
text(0.5,0.65,
     paste("Highest Risk:",highest),
     cex=1.1)
text(0.5,0.50,
     paste("Age:",patient$Age[
       which.max(patient$Risk_Score)]),
     cex=1.1)
text(0.5,0.35,
     paste("Highest BP:",max(patient$BP)),
     cex=1.1)
text(0.5,0.20,

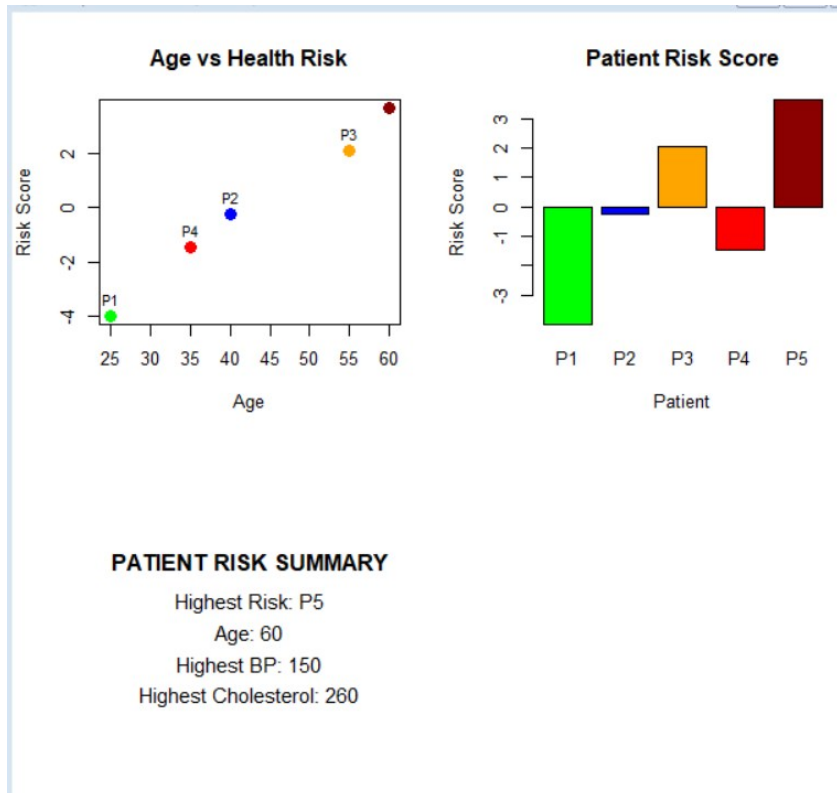
```

```

paste("Highest Cholesterol:",
      max(patient$Cholesterol)),
cex=1.1)
par(mfrow=c(1,1))

```

OUTPUT:



SET 28 – Vehicle Performance Analysis

Q1. Violin Plot – Fuel Efficiency by Safety Rating

CODE:

```

install.packages("ggplot2")
library(ggplot2)
vehicle <- data.frame(
  Safety_Rating = factor(c(4,5,5,4,3)),
  Fuel_Efficiency = c(18,15,12,14,17)
)
ggplot(vehicle,
  aes(x=Safety_Rating,
      y=Fuel_Efficiency,

```

```

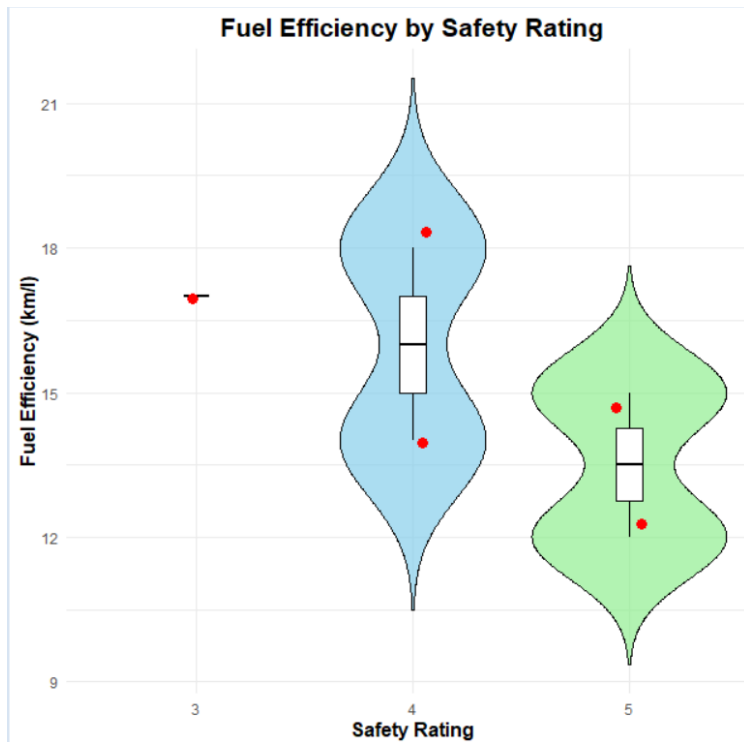
    fill=Safety_Rating)) +
geom_violin(trim=FALSE,
    alpha=0.7,
    color="black") +
geom_boxplot(width=0.12,
    fill="white",
    color="black") +
geom_jitter(width=0.08,
    size=3,
    color="red") +
labs(
    title="Fuel Efficiency by Safety Rating",
    x="Safety Rating",
    y="Fuel Efficiency (km/l)"
) +
scale_fill_manual(values=c(
    "3"="orange",
    "4"="skyblue",
    "5"="lightgreen"
)) +
theme_minimal() +
theme(
    plot.title=element_text(
        size=16,
        face="bold",
        hjust=0.5
    ),
    axis.title=element_text(
        size=12,
        face="bold"
    ),

```

```
legend.position="none"
```

```
)
```

OUTPUT:



Q2. Horsepower vs Top Speed

CODE:

```
engine <- c(1.5,2.0,3.0,2.5,1.8)
horsepower <- c(110,150,250,200,130)
top_speed <- c(180,200,250,220,190)
colour <- c("blue","green","red","orange","purple")
plot(horsepower,
     top_speed,
     pch=19,
     cex=1.7,
     col=colour,
     xlab="Horsepower",
     ylab="Top Speed (km/h)",
     main="Horsepower vs Top Speed")
abline(lm(top_speed ~ horsepower),
```

```

col="darkblue",

lwd=3)

legend("topleft",

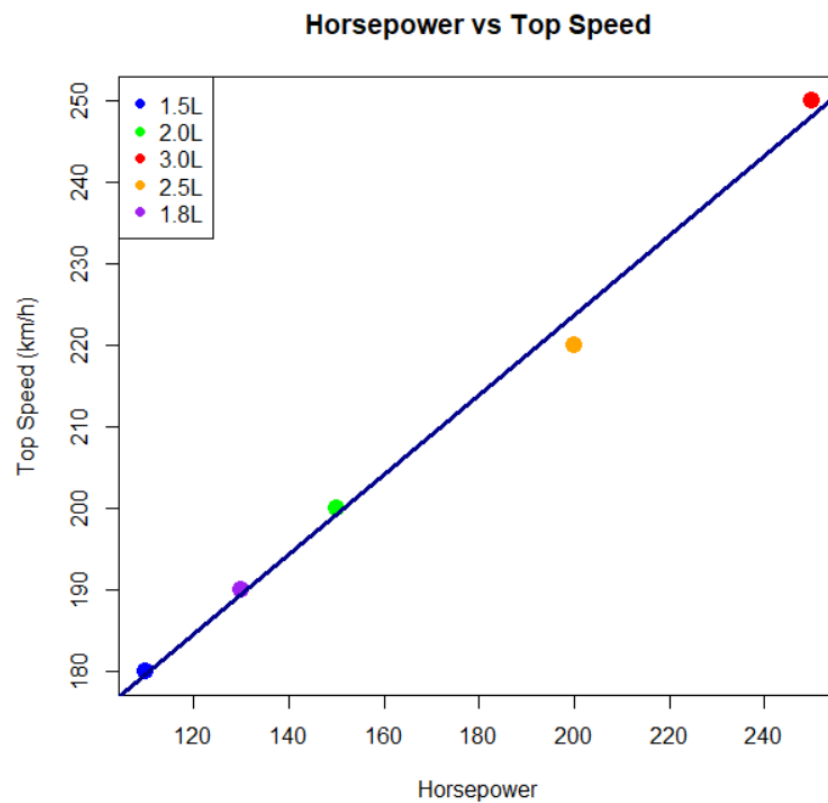
legend=c("1.5L","2.0L","3.0L","2.5L","1.8L"),

col=colour,

pch=19)

```

OUTPUT:



Q3. Correlation Heatmap

CODE:

```

vehicle <- data.frame(

  Engine_Size = c(1.5,2.0,3.0,2.5,1.8),

  Horsepower = c(110,150,250,200,130),

  Fuel_Efficiency = c(18,15,12,14,17),

  Top_Speed = c(180,200,250,220,190),

  Safety_Rating = c(4,5,5,4,3)

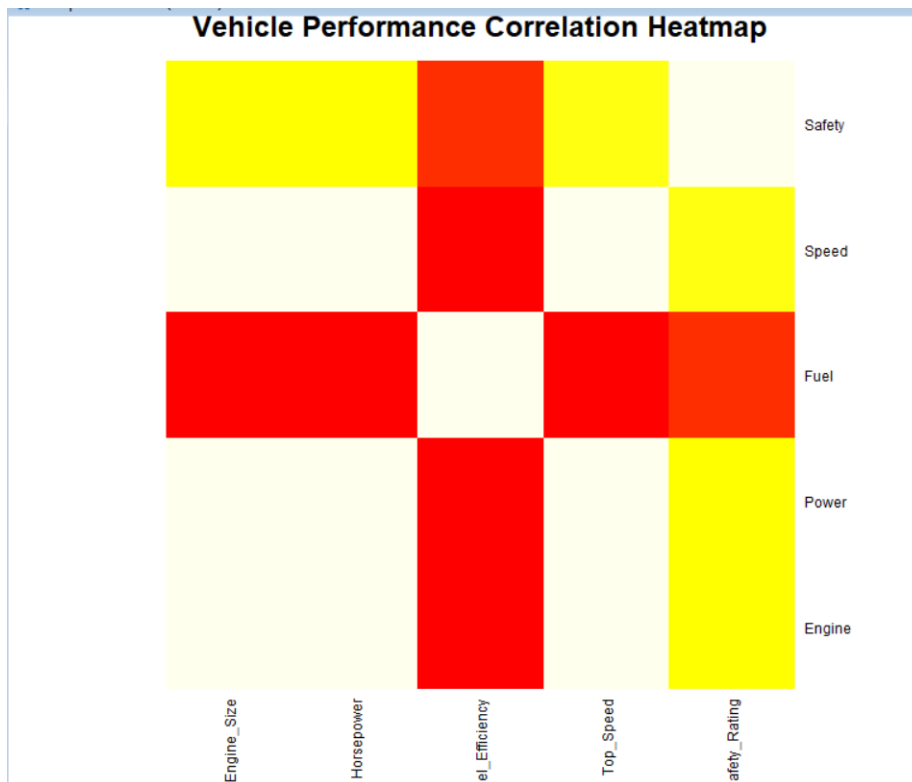
)

```

```
cor_matrix <- cor(vehicle)

heatmap(cor_matrix,
        Rowv=NA,
        Colv=NA,
        scale="none",
        col=heat.colors(20),
        main="Vehicle Performance Correlation Heatmap")
```

OUTPUT:



Q4. Vehicle Performance Dashboard in R

CODE:

```
vehicle <- data.frame(
  Vehicle_ID=c("V1","V2","V3","V4","V5"),
  Engine_Size=c(1.5,2.0,3.0,2.5,1.8),
  Horsepower=c(110,150,250,200,130),
  Fuel_Efficiency=c(18,15,12,14,17),
  Top_Speed=c(180,200,250,220,190),
  Safety_Rating=c(4,5,5,4,3)
)
```



```

par(mfrow=c(2,2),
    mar=c(5,4,4,2))
barplot(vehicle$Fuel_Efficiency,
        names.arg=vehicle$Vehicle_ID,
        col=c("green","blue","red","orange","purple"),
        main="Fuel Efficiency",
        xlab="Vehicle",
        ylab="km/l")
plot(vehicle$Horsepower,
      vehicle$Top_Speed,
      pch=19,
      cex=1.6,
      col=c("blue","green","red","orange","purple"),
      xlab="Horsepower",
      ylab="Top Speed (km/h)",
      main="Horsepower vs Top Speed")
abline(lm(Top_Speed ~ Horsepower,
          data=vehicle),
       col="darkblue",
       lwd=3)
barplot(vehicle$Safety_Rating,
        names.arg=vehicle$Vehicle_ID,
        col=c("lightblue","lightgreen",
              "orange","pink","yellow"),
        main="Safety Rating",
        xlab="Vehicle",
        ylab="Rating")
plot.new()
text(0.5,0.85,
     "VEHICLE PERFORMANCE SUMMARY",
     cex=1.3,

```

```

font=2)

best_speed <- vehicle$Vehicle_ID[
  which.max(vehicle$Top_Speed)]

best_efficiency <- vehicle$Vehicle_ID[
  which.max(vehicle$Fuel_Efficiency)]

text(0.5,0.65,
     paste("Highest Speed:",best_speed),
     cex=1.1)

text(0.5,0.50,
     paste("Top Speed:",
           max(vehicle$Top_Speed),"km/h"),
     cex=1.1)

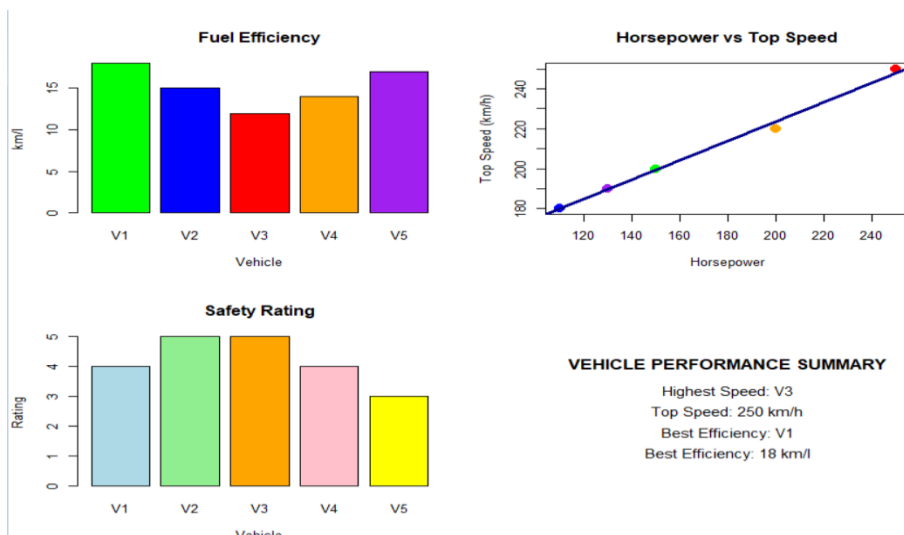
text(0.5,0.35,
     paste("Best Efficiency:",
           best_efficiency),
     cex=1.1)

text(0.5,0.20,
     paste("Best Efficiency:",
           max(vehicle$Fuel_Efficiency),"km/l"),
     cex=1.1)

par(mfrow=c(1,1))

```

OUTPUT:



SET 29 – Student Academic Performance

Q1. Stacked Area Chart – Test Score & Participation Score

CODE:

```
student <- data.frame(  
  Student_ID = c("S1", "S2", "S3", "S4", "S5"),  
  Test_Score = c(85, 70, 92, 80, 65),  
  Participation_Score = c(8, 7, 9, 8, 6)  
)  
  
data <- rbind(  
  Test_Score = student$Test_Score,  
  Participation_Score = student$Participation_Score  
)  
  
plot(1:5,  
  student$Test_Score,  
  type="n",  
  ylim=c(0, 110),  
  xaxt="n",  
  xlab="Student",  
  ylab="Score",  
  main="Test and Participation Scores")  
  
polygon(c(1:5, 5:1),  
  c(student$Test_Score, rep(0, 5)),  
  col="skyblue")  
  
polygon(c(1:5, 5:1),  
  c(student$Test_Score,  
    rev(student$Test_Score +  
      student$Participation_Score)),  
  col="orange")  
  
axis(1,  
  at=1:5,  
  labels=student$Student_ID)
```

```

legend("topleft",
      legend=c("Test Score", "Participation Score"),
      fill=c("skyblue", "orange"))

```

OUTPUT:



Q2. Boxplot – Study Hours by Attendance Quartile

CODE:

```

student <- data.frame(
  Study_Hours = c(12,8,15,10,7),
  Attendance = c(90,70,95,85,60)
)

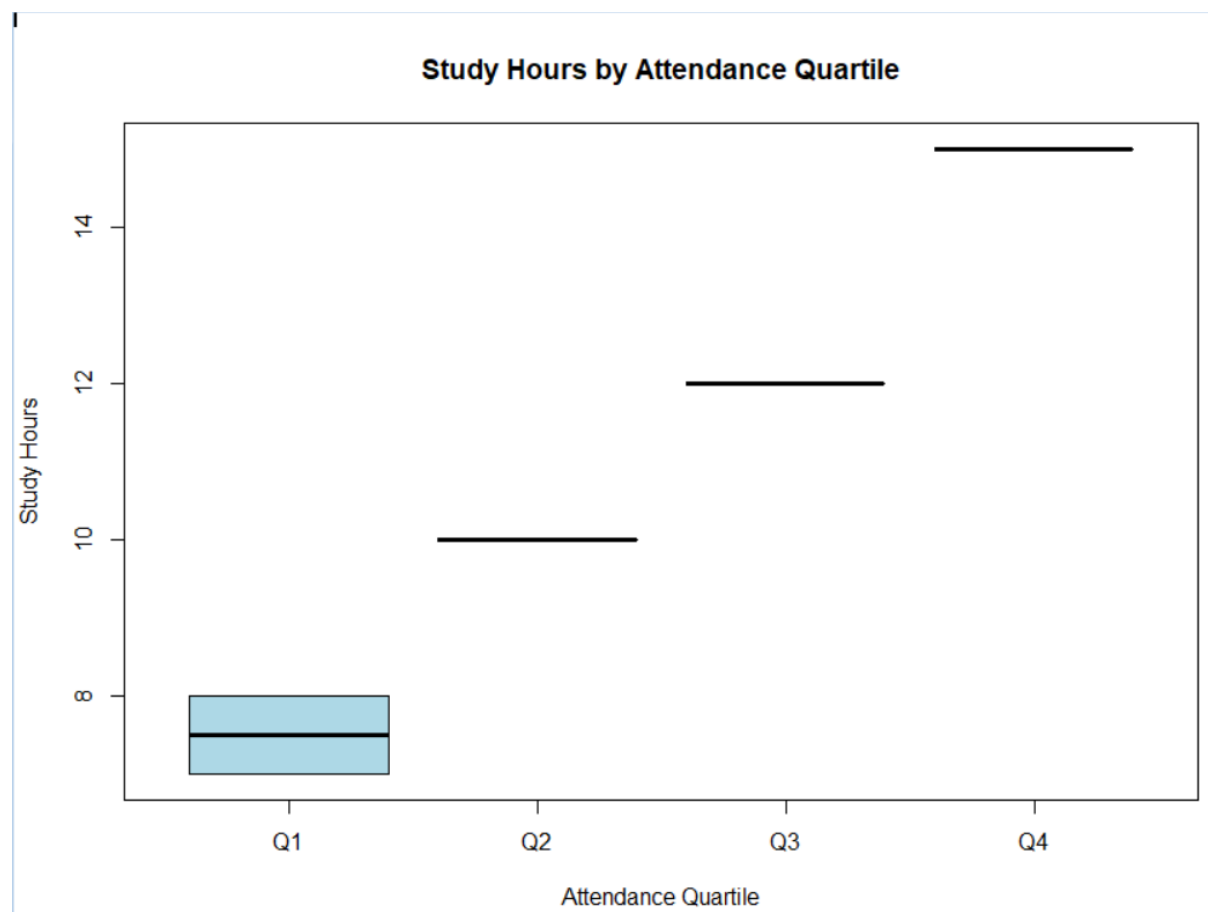
student$Attendance_Quartile <- cut(
  student$Attendance,
  breaks=quantile(student$Attendance,
    probs=c(0,0.25,0.5,0.75,1)),
  include.lowest=TRUE,
  labels=c("Q1","Q2","Q3","Q4")
)

```

)

```
boxplot(Study_Hours ~ Attendance_Quartile,  
        data=student,  
        col=c("lightblue",  
              "lightgreen",  
              "orange",  
              "pink"),  
        main="Study Hours by Attendance Quartile",  
        xlab="Attendance Quartile",  
        ylab="Study Hours",  
        border="black")
```

OUTPUT:



Q3. Density Plot – Test Score

CODE:

```
student <- data.frame(
```

```
Test_Score = c(85,70,92,80,65)
)
d <- density(student$Test_Score)
plot(d,
      col="blue",
      lwd=3,
      main="Density Plot of Test Scores",
      xlab="Test Score",
      ylab="Density")
polygon(d,
        col="yellow",
        border="green")
rug(student$Test_Score,
     col="red",
     lwd=2)
```

output:

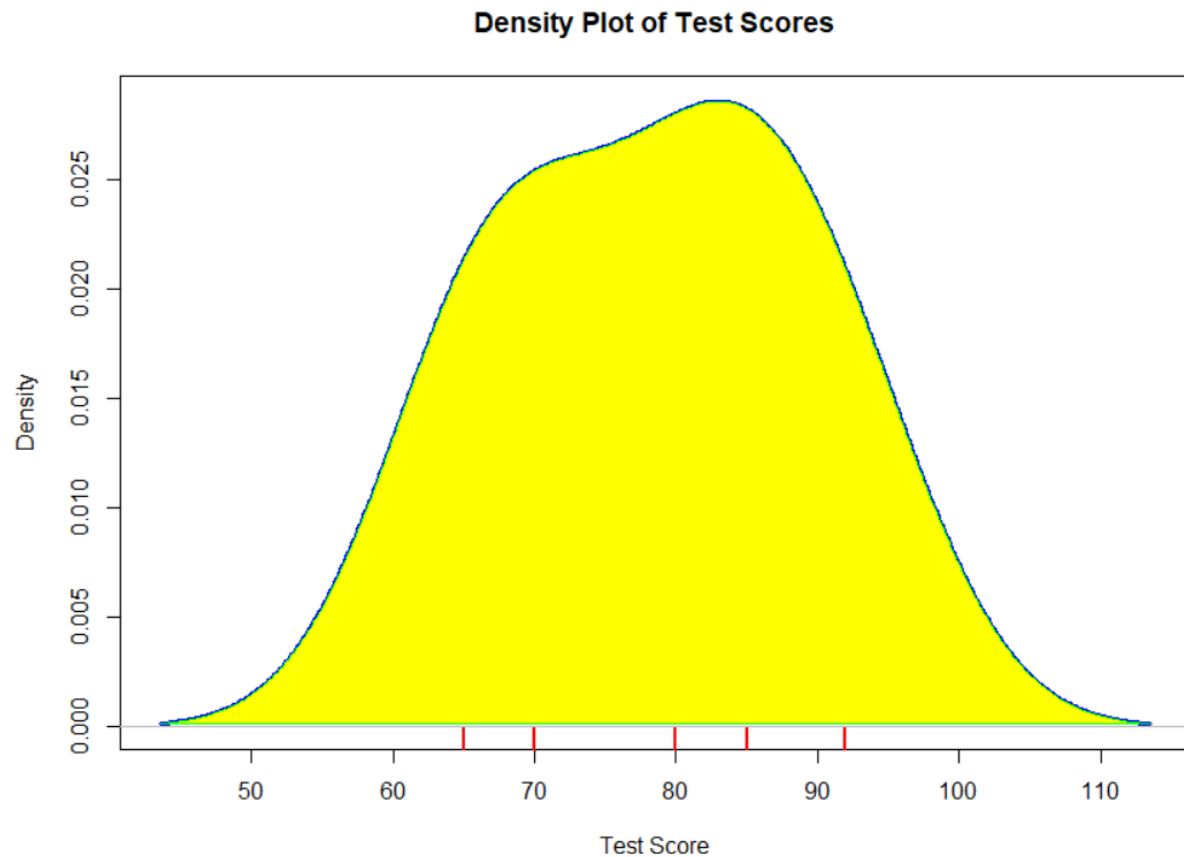
```
student <- data.frame(
  Test_Score = c(85,70,92,80,65)
)
d <- density(student$Test_Score)
plot(d,
      col="blue",
      lwd=3,
      main="Density Plot of Test Scores",
      xlab="Test Score",
      ylab="Density")

polygon(d,
        col="yellow",
        border="green")
rug(student$Test_Score,
```

```
col="red",
```

```
lwd=2)
```

OUTPUT:



Q4. R Dashboard – Study Hours vs Test Score

CODE:

```
student <- data.frame(  
  Student_ID=c("S1","S2","S3","S4","S5"),  
  Age=c(19,21,20,22,23),  
  Study_Hours=c(12,8,15,10,7),  
  Attendance=c(90,70,95,85,60),  
  Test_Score=c(85,70,92,80,65),  
  Participation_Score=c(8,7,9,8,6)  
)  
par(mfrow=c(2,2),  
    mar=c(5,4,4,2))  
plot(student$Study_Hours,
```

```
student$Test_Score,
pch=19,
cex=1.8,
col=c("blue","green","red","orange","purple"),
xlab="Study Hours",
ylab="Test Score",
main="Study Hours vs Test Score")
text(student$Study_Hours,
student$Test_Score,
labels=student$Student_ID,
pos=3)
abline(lm(Test_Score ~ Study_Hours,
data=student),
col="darkblue",
lwd=3)
barplot(student$Participation_Score,
names.arg=student$Student_ID,
col=c("blue","green","red","orange","purple"),
main="Participation Score",
xlab="Student",
ylab="Participation Score")
barplot(student$Attendance,
names.arg=student$Student_ID,
col="skyblue",
main="Student Attendance",
xlab="Student",
ylab="Attendance (%)")
plot.new()
text(0.5,0.85,
"ACADEMIC PERFORMANCE SUMMARY",
cex=1.2,
```

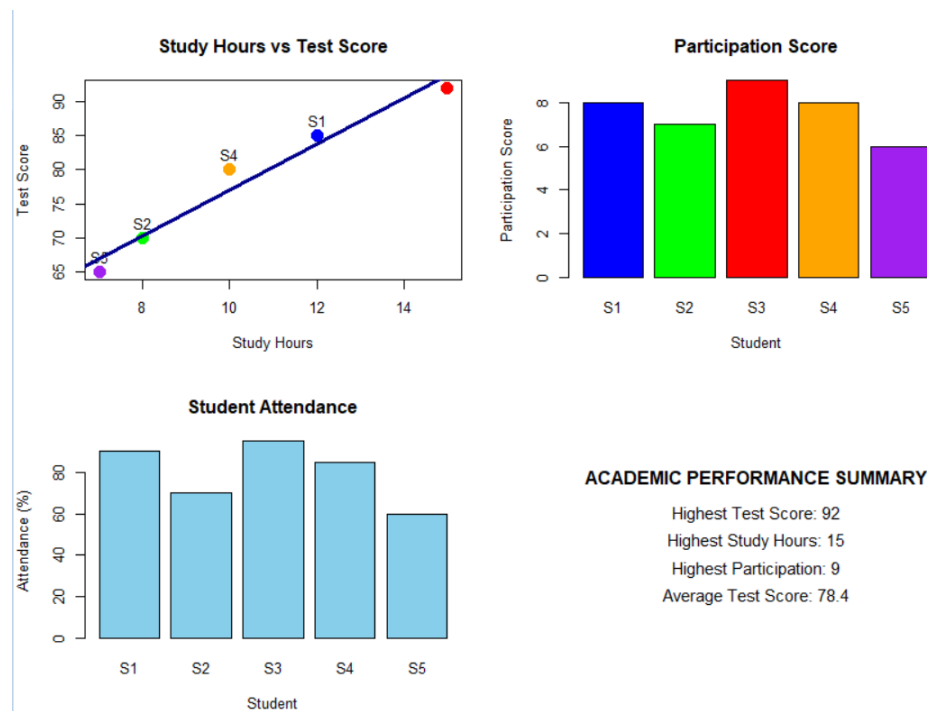


```

font=2)
text(0.5,0.65,
     paste("Highest Test Score:",
           max(student$Test_Score)),
     cex=1.1)
text(0.5,0.50,
     paste("Highest Study Hours:",
           max(student$Study_Hours)),
     cex=1.1)
text(0.5,0.35,
     paste("Highest Participation:",
           max(student$Participation_Score)),
     cex=1.1)
text(0.5,0.20,
     paste("Average Test Score:",
           round(mean(student$Test_Score),1)),
     cex=1.1)
par(mfrow=c(1,1))

```

OUTPUT:



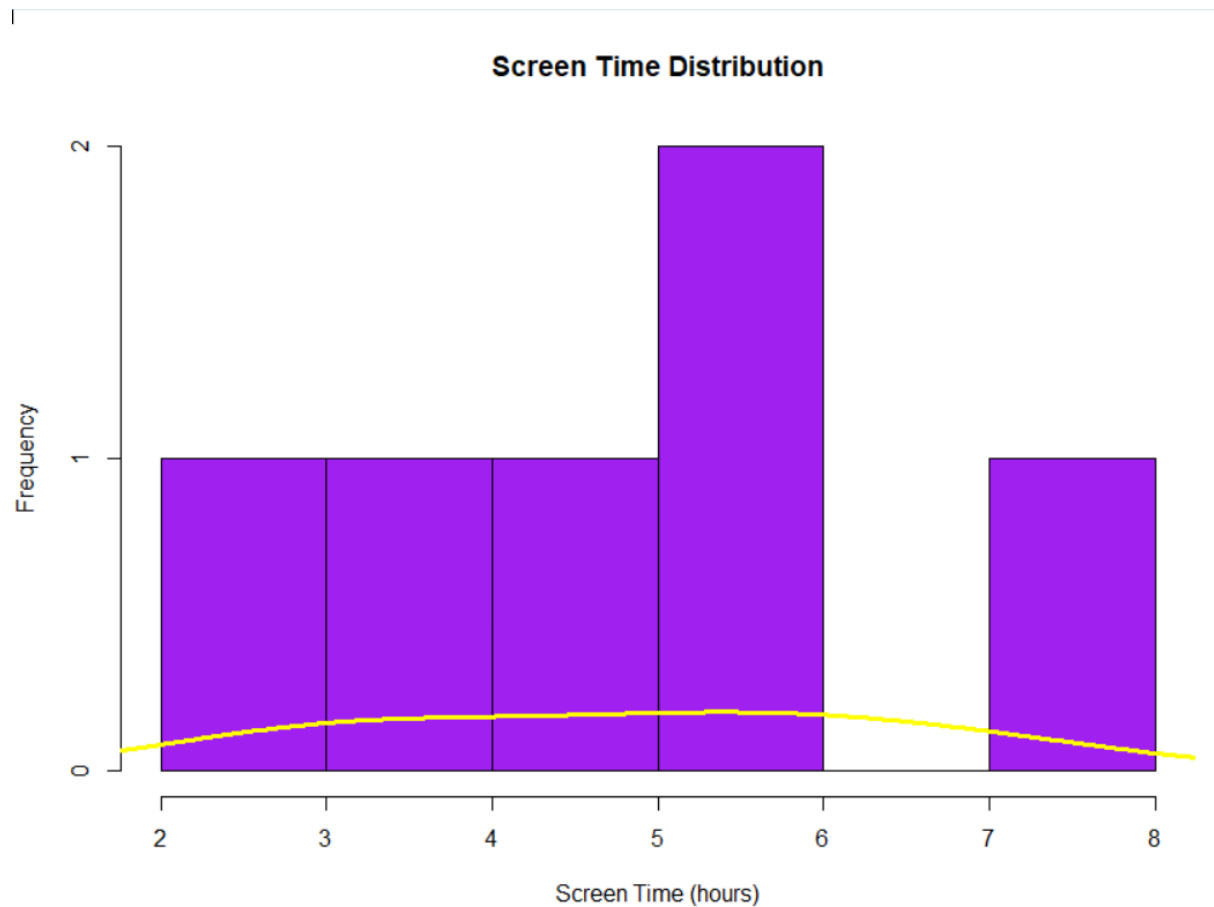
SET 30 – Mobile App Usage Analysis

Q1. Histogram + Density Plot for Screen Time

CODE:

```
screen_time <- c(4.5,6.0,3.2,7.1,2.8,5.4)
hist(screen_time,
      breaks=5,
      col="purple",
      border="black",
      main="Screen Time Distribution",
      xlab="Screen Time (hours)",
      ylab="Frequency")
lines(density(screen_time),
      col="yellow",
      lwd=3)
```

OUTPUT:

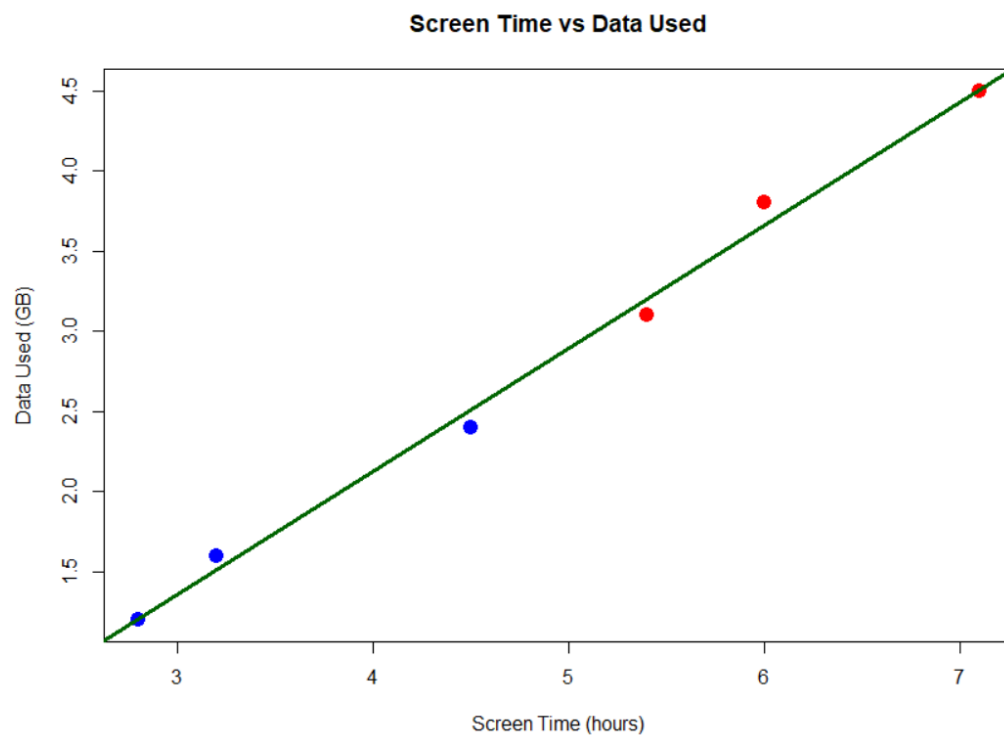


Q2. Data Used vs Screen Time

CODE:

```
screen_time <- c(4.5,6.0,3.2,7.1,2.8,5.4)
data_used <- c(2.4,3.8,1.6,4.5,1.2,3.1)
plot(screen_time,
      data_used,
      pch=19,
      cex=1.6,
      col=c("blue","red","blue","red","blue","red"),
      main="Screen Time vs Data Used",
      xlab="Screen Time (hours)",
      ylab="Data Used (GB)")
abline(lm(data_used ~ screen_time),
      col="darkgreen",
      lwd=3)
correlation <- cor(screen_time,data_used)
print(paste("Correlation =",round(correlation,2)))
```

OUTPUT:

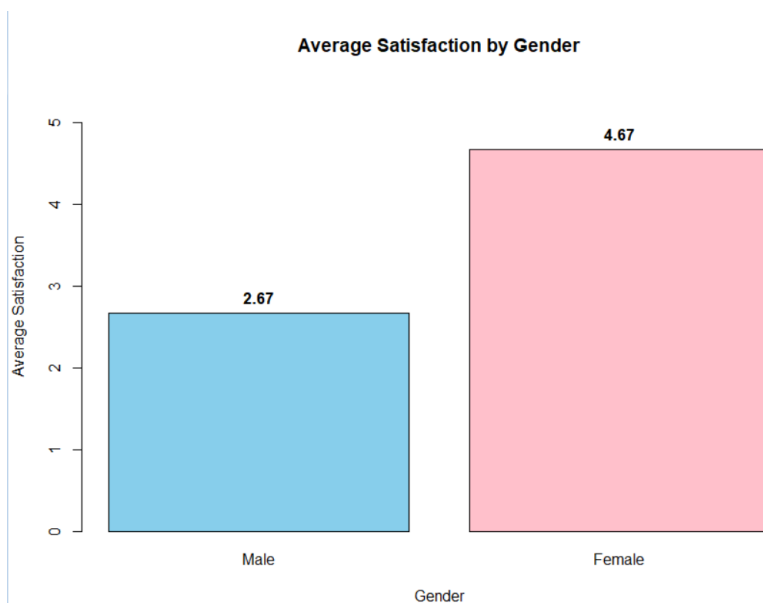


Q3. Average Satisfaction by Gender

CODE:

```
gender <- c("Male","Female","Male","Female","Male","Female")
satisfaction <- c(3,5,3,5,2,4)
male_avg <- mean(satisfaction[gender=="Male"])
female_avg <- mean(satisfaction[gender=="Female"])
average <- c(male_avg,female_avg)
bars <- barplot(average,
                 names.arg=c("Male","Female"),
                 col=c("skyblue","pink"),
                 main="Average Satisfaction by Gender",
                 xlab="Gender",
                 ylab="Average Satisfaction",
                 ylim=c(0,5.5))
text(bars,
     average,
     labels=round(average,2),
     pos=3,
     cex=1.1,
     font=2)
```

OUTPUT:



Q4. Mini Dashboard in R

CODE:

```
mobile <- data.frame(
  User_ID=c("U01","U02","U03","U04","U05","U06"),
  Gender=c("Male","Female","Male","Female","Male","Female"),
  Age=c(20,22,19,21,23,20),
  Screen_Time=c(4.5,6.0,3.2,7.1,2.8,5.4),
  App_Usage_Count=c(18,25,12,30,10,22),
  Data_Used=c(2.4,3.8,1.6,4.5,1.2,3.1),
  Satisfaction=c(3,5,3,5,2,4),
  Usage_Date=as.Date(c("2025-01-08",
                        "2025-01-08",
                        "2025-02-11",
                        "2025-02-11",
                        "2025-03-14",
                        "2025-03-14"))
)

par(mfrow=c(2,2),
    mar=c(5,4,4,2))
plot(mobile$Screen_Time,
     mobile$Data_Used,
     pch=21,
     bg=ifelse(mobile$Gender=="Male",
               "skyblue","pink"),
     col="black",
     cex=mobile$App_Usage_Count/8,
     main="Screen Time vs Data Used",
     xlab="Screen Time (hours)",
     ylab="Data Used (GB)")
text(mobile$Screen_Time,
```

```

mobile$Data_Used,
labels=mobile$User_ID,
pos=3,
cex=0.8)
legend("topleft",
      legend=c("Male","Female"),
      pt.bg=c("skyblue","pink"),
      pch=21,
      title="Gender")
plot(mobile$Usage_Date,
      mobile$Screen_Time,
      type="o",
      pch=19,
      col="darkblue",
      main="Screen Time Trend",
      xlab="Usage Date",
      ylab="Screen Time (hours)")
barplot(mobile$App_Usage_Count,
        names.arg=mobile$User_ID,
        col=c("blue","pink",
              "blue","pink",
              "blue","pink"),
        main="App Usage Count",
        xlab="User",
        ylab="Usage Count")
plot.new()
text(0.5,0.85,
     "MOBILE APP USAGE SUMMARY",
     cex=1.3,
     font=2)
text(0.5,0.65,

```

```

paste("Average Screen Time:",
      round(mean(mobile$Screen_Time),2),
      "hrs"),
cex=1.1)
text(0.5,0.50,
     paste("Average Data Used:",
           round(mean(mobile$Data_Used),2),
           "GB"),
     cex=1.1)
text(0.5,0.35,
     paste("Highest Screen Time:",
           max(mobile$Screen_Time),
           "hrs"),
     cex=1.1)
text(0.5,0.20,
     paste("Highest Satisfaction:",
           max(mobile$Satisfaction)),
     cex=1.1)
par(mfrow=c(1,1))

```

OUTPUT:

